Incidence-Guided Treatment Assignment Designs for Spatial Cluster Randomized Trials

Comprehensive Simulation Study Report

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1 Introduction

This report provides a comprehensive reference for the simulation study evaluating **8 treatment assignment designs** for Spatial Cluster Randomized Trials (CRTs) when outcome incidence is heterogeneous and spatially structured.

> *Given known historical incidence data exhibiting spatial heterogeneity, which treatment assignment strategy minimizes estimation error for the direct intervention effect $\hat{\tau}$ across a range of spatial dependence and spillover configurations?*

Application context: Sudden Unexpected Death (SUD) in North Carolina counties. The Poisson incidence mode uses an SUD base rate of 35 per 100,000, consistent with county-level analyses.

The study uses Maximum Likelihood Estimation via a spatial autoregressive model (MLE) as the primary estimator, evaluated across 2,560 parameter scenarios. Three analytical questions structure the results:

- **Q1:** Which design performs best within each incidence generation mode?
- **Q2:** How do design rankings change as individual parameters vary?
- **Q3:** Which design wins for specific parameter combinations?

---

2 Study Region and Spatial Structure

The study region is a regular square lattice grid:

$$\mathcal{G} = \{s_1, s_2, \dots, s_N\}$$

where each cluster s_i is located at integer coordinates (x_i, y_i) on a $G \times G$ grid, so $N = G^2$. In this study, $G = 10$ and $N = 100$.

2.1 Spatial Contiguity

Two contiguity definitions are considered:

- **Rook contiguity**: clusters sharing an edge (4-connected).

$$\mathcal{N}_R(s_i) = \{s_j : |x_i - x_j| + |y_i - y_j| = 1\}$$

- **Queen contiguity**: clusters sharing an edge or vertex (8-connected).

$$\mathcal{N}_Q(s_i) = \{s_j : \max(|x_i - x_j|, |y_i - y_j|) = 1\}$$

2.2 Spatial Weight Matrix

Let $\mathbf{A}$ denote the binary adjacency matrix where $A_{ij} = \mathbf{1}[s_j \in \mathcal{N}(s_i)]$. The row-standardized spatial weight matrix is:

$$W_{ij} = \frac{A_{ij}}{\sum_{k=1}^{N} A_{ik}}, \quad i \neq j; \quad W_{ii} = 0$$

This ensures $\sum_j W_{ij} = 1$ for each row, so spatial lags represent neighborhood averages. We construct $\mathbf{W}_R$ (rook) and $\mathbf{W}_Q$ (queen) weight matrices.

---

3 Outcome Data Generating Process

3.1 Spatial Durbin Model (SDM)

The outcome vector $\mathbf{Y} = (Y_1, \dots, Y_N)^\top$ is generated from the Spatial Durbin Model:

$$\mathbf{Y} = (\mathbf{I}_N - \rho\mathbf{W})^{-1} \left[\tau\mathbf{Z} + \gamma \cdot \text{Spill}(\mathbf{Z}) + \beta\mathbf{X} + \varepsilon\right]$$

where:

Symbol	Description	Role
$\mathbf{Y}$	$N \times 1$ outcome vector	Response
ρ	Spatial autocorrelation parameter	Controls global spatial dependence in $\mathbf{Y}$
$\mathbf{W}$	$N \times N$ row-standardized weight matrix	Spatial structure (rook or queen)
τ	Direct treatment effect	**Estimand of interest**
$\mathbf{Z}$	$N \times 1$ binary treatment vector	$Z_i \in \{0, 1\}$

Symbol	Description	Role
γ	Spillover effect magnitude	Indirect effect via treated neighbors
$\text{Spill}(\mathbf{Z})$	$N \times 1$ spillover term	See below
β	Effect of baseline incidence on outcome	Covariate coefficient
$\mathbf{X}$	$N \times 1$ baseline incidence vector	$X_i \in [0, 1]$
ε	$N \times 1$ error vector	$\varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$

The spatial multiplier $(\mathbf{I}_N - \rho\mathbf{W})^{-1}$ induces global spatial dependence.

Stationarity condition: $|\rho| < 1/\lambda_{\max}(\mathbf{W})$. For row-standardized $\mathbf{W}$, $\lambda_{\max} = 1$, so $|\rho| < 1$.

3.2 Spillover Construction

$$\text{Spill}_i = \frac{\sum_{j \in \mathcal{N}(i)} Z_j}{|\mathcal{N}(i)|} = (\mathbf{WZ})_i$$

Two spillover types are considered:

1. **Control-only spillover**: $\text{Spill}_i^{(\text{ctrl})} = \gamma \cdot (\mathbf{WZ})_i \cdot (1 - Z_i)$
2. **Bidirectional spillover**: $\text{Spill}_i^{(\text{both})} = \gamma \cdot (\mathbf{WZ})_i$

---

4 Incidence Generation Modes

The baseline incidence vector $\mathbf{X}$ represents prior observed outcome rates. Three generation modes are considered, all producing values in $[0, 1]$.

4.1 Mode 1: Independent and Identically Distributed (iid)

$$X_i \overset{\text{iid}}{\sim} \text{Uniform}(0, 1), \quad i = 1, \dots, N$$

No spatial structure; serves as the baseline. One config ($\rho_X = \text{N/A}$).

4.2 Mode 2: Spatially Correlated (SAR Filter)

$$\mathbf{X}^* = (\mathbf{I}_N - \rho_X \mathbf{W})^{-1}\mathbf{u}, \quad \mathbf{u} \sim N(\mathbf{0}, \mathbf{I}_N)$$

$$X_i = \Phi(X_i^*)$$

where $\Phi(\cdot)$ is the standard normal CDF. Two configs: $\rho_X \in \{0.20, 0.50\}$.

4.3 Mode 3: Poisson Count-Based

1. Spatially correlated log-rates: $\log \lambda^* = \mu_0 \cdot \mathbf{1} + (\mathbf{I}_N - \rho_X \mathbf{W})^{-1}\mathbf{u}$
2. Poisson counts: $C_i \sim \text{Poisson}(\lambda_i^* \cdot P_i)$ where $P_i = 1{,}000$
3. Observed rates: $R_i = C_i / P_i$
4. Rank-normalized: $X_i = (\text{rank}(R_i) - 0.5)/N$

Base rate $\mu_0 = \log(35/100{,}000)$ for SUD. Two configs: $\rho_X \in \{0.20, 0.50\}$.

Critical design invariant: Incidence is generated **once per (mode, ρ_X) configuration** and held fixed across all (ρ, γ, nb_type) scenarios.

4.4 Incidence Spatial Patterns

---

5 Treatment Assignment Designs

Eight designs are evaluated. Each produces a binary treatment vector $\mathbf{Z} \in \{0, 1\}^N$.

Table 2: 8 treatment assignment strategies evaluated in this simulation

ID	Name	Deterministic?
1	Design 1: Checkerboard	Yes
2	Design 2: High Incidence Focus	Yes
3	Design 3: Saturation Quadrants	No
4	Design 4: Isolation Buffer	No
5	Design 5: 2x2 Blocking	No
6	Design 6: Balanced Quartiles	No
7	Design 7: Balanced Halves	No
8	Design 8: Incidence-Guided Saturation Quadrants	No

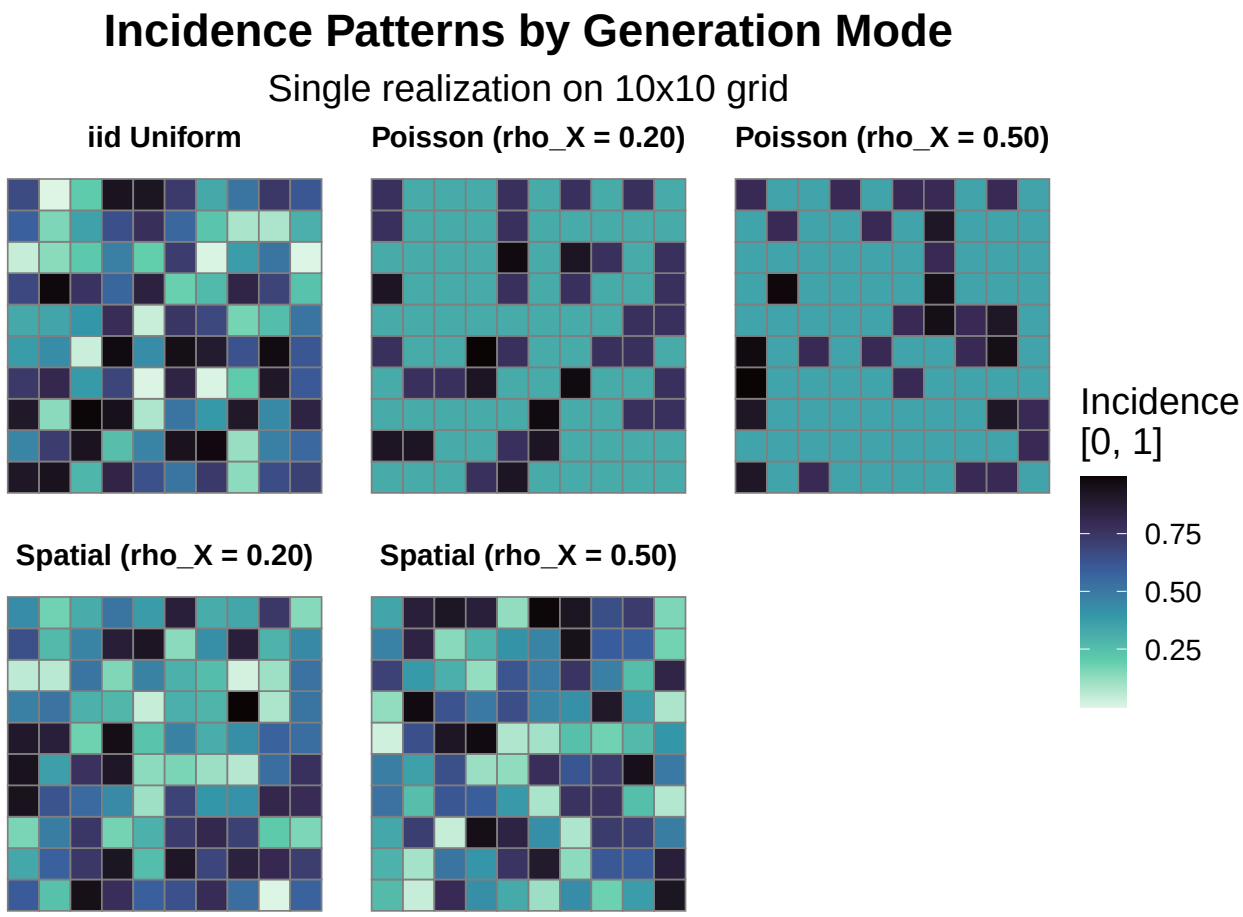

Figure 1: Incidence spatial patterns across generation modes. Color represents incidence (darker = higher burden). Spatial and Poisson modes show visible spatial clustering absent from iid.

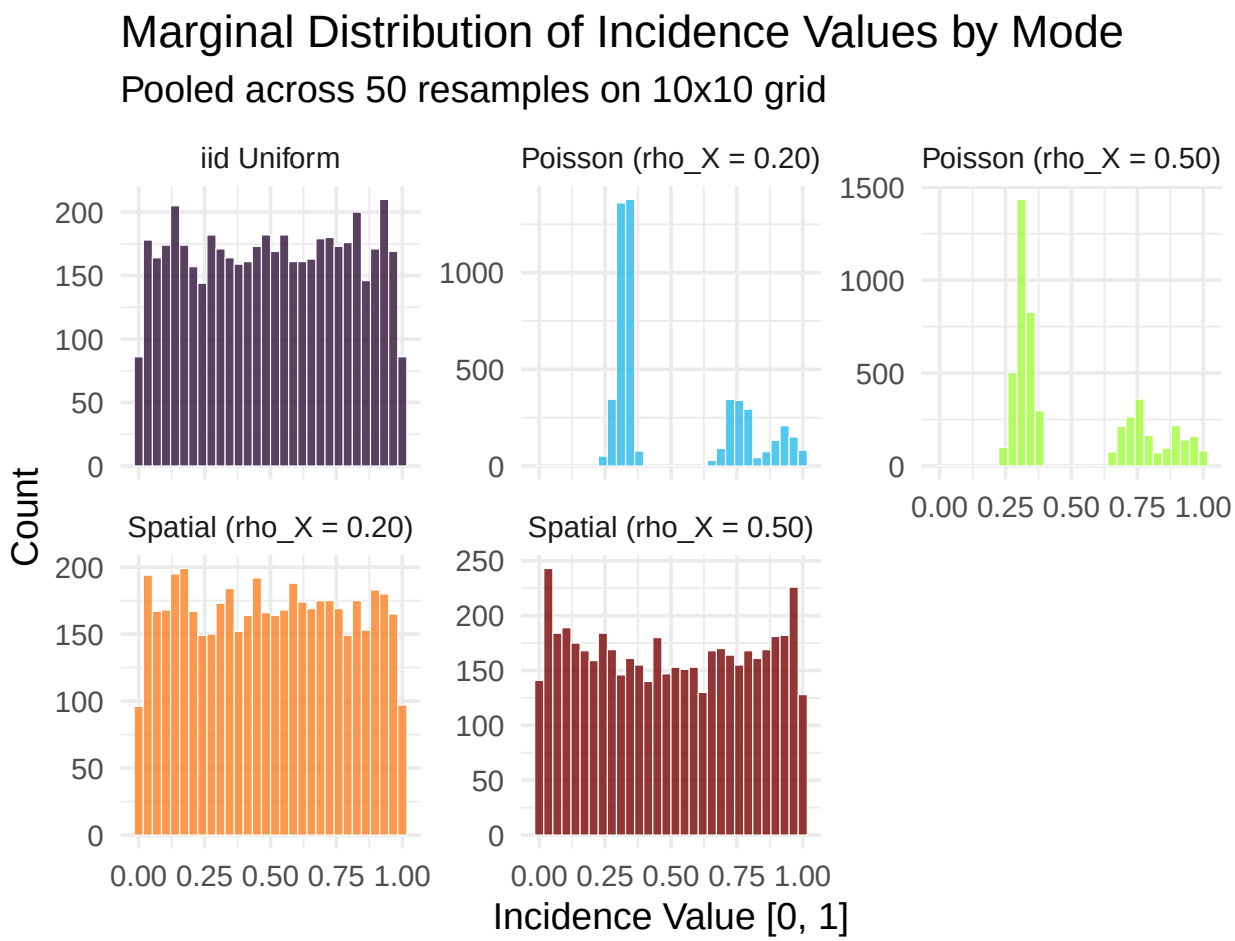

Figure 2: Marginal distribution of incidence values by mode, pooled across 50 resamples. All modes are marginally Uniform on [0,1]; spatial structure manifests in spatial covariance, not the marginal.

5.1 Design 1: Checkerboard (Strict Spatial Stratification)

$$Z_i = (x_i + y_i) \mod 2$$

Treatment in an alternating checkerboard pattern. **Deterministic**, exactly 50% treated.

5.2 Design 2: High Incidence Focus

$$Z_i = \mathbf{1}[X_i > \text{median}(\mathbf{X})]$$

Treatment assigned to above-median incidence clusters. **Deterministic** given $\mathbf{X}$.

5.3 Design 3: Saturation Quadrants

Grid partitioned into four spatial quadrants. Saturation levels $\{0.20, 0.40, 0.60, 0.80\}$ randomly assigned to quadrants. Within each quadrant, clusters randomly selected at the assigned rate.

5.4 Design 4: Isolation Buffer

Sequential treatment assignment: randomly select an available cluster, assign treatment, remove it and its neighbors from the available set. Treated clusters are never adjacent.

5.5 Design 5: 2x2 Blocking

Grid partitioned into non-overlapping 2×2 blocks. Within each block, exactly 2 of 4 clusters randomly assigned to treatment.

5.6 Design 6: Balanced Quartiles

Clusters stratified by incidence quartile. Within each stratum, approximately half randomly assigned to treatment.

5.7 Design 7: Balanced Halves

Clusters stratified into two groups by median baseline incidence. Within each half, approximately half randomly assigned to treatment. A 2-stratum analog of Design 6.

5.8 Design 8: Incidence-Guided Saturation Quadrants

Grid partitioned into four spatial quadrants (same as Design 3). Quadrants ranked by average baseline incidence and **deterministically** assigned saturation levels:

Rank (by $\bar{X}_{Q_k}$)	Saturation p_k
1 (highest)	0.80
2	0.60
3	0.40
4 (lowest)	0.20

Within each quadrant, clusters randomly selected at the assigned rate. Unlike Design 3, the saturation-to-quadrant mapping is guided by incidence data rather than random.

---

6 Estimation Methods

6.1 Difference in Means (DIM)

The naive estimator:

$$\hat{\tau}_{\text{DIM}} = \bar{Y}_1 - \bar{Y}_0$$

DIM is biased for τ when spillover ($\gamma \neq 0$) or spatial dependence ($\rho \neq 0$) is present.

6.2 Maximum Likelihood Estimation (MLE)

The spatial autoregressive model is estimated via maximum likelihood using `spatialreg::lagsarlm()`:

$$Y_i = \rho \sum_j W_{ij} Y_j + \tau Z_i + \gamma \cdot \text{Spill}_i + \beta X_i + \varepsilon_i$$

MLE simultaneously estimates ρ and regression coefficients. The true spillover covariate Spill($\mathbf{Z}$) is included as an **oracle regressor**. Confidence intervals for τ use the asymptotic covariance matrix.

MLE is the primary estimator throughout this report. DIM serves only as a naive baseline.

---

7 Simulation Design

7.1 Parameter Grid

Table 4: Simulation parameter grid (2,560 total scenarios)

Parameter	Values	Levels
Incidence mode	iid, poisson, spatial	5 configs (1 iid + 2 spatial + 2 poisson)
rho_X (incidence spatial autocorr.)	N/A (iid); 0.20, 0.50 (spatial, poisson)	—
nb_type (contiguity)	queen, rook	2
rho (outcome spatial autocorr.)	0, 0.01, 0.2, 0.5	4
gamma (spillover magnitude)	0.5, 0.6, 0.7, 0.8	4
spill_type	both, control_only	2
design_id	1, 2, 3, 4, 5, 6, 7, 8	8
TOTAL		2,560

7.2 Resampling Structure

Mode	Design resamples (R_d)	Outcome resamples (R_o)	Iterations / scenario
MLE	25	10	250

7.3 Per-Scenario Reproducibility

Each scenario uses a deterministic seed:

$$\text{seed}_s = \texttt{digest::digest2int}(\texttt{paste}(\text{inc_mode}, \rho_X, \text{nb_type}, \rho, \gamma, \text{spill_type}, d_{\text{id}}, \text{sep} = \text{"|"}))$$

This ensures results are reproducible and order-independent.

---

8 Evaluation Metrics

For each scenario, let $\hat{\tau}_1, \ldots, \hat{\tau}_K$ denote the K estimates of τ (excluding convergence failures).

- **Bias**: $\text{Bias} = \bar{\hat{\tau}} - \tau$
- **Standard Deviation**: $\text{SD} = \sqrt{\frac{1}{K-1} \sum_{k=1}^{K} \left(\hat{\tau}_k - \bar{\hat{\tau}}\right)^2}$
- **Mean Squared Error**: $\text{MSE} = \text{Bias}^2 + \text{SD}^2$ — **primary ranking criterion**
- **Coverage Probability**: $\text{Coverage} = \frac{1}{K} \sum_{k=1}^{K} \mathbf{1}\left[\tau \in \left(\hat{\tau}_{k,L},\ \hat{\tau}_{k,U}\right)\right]$
- **Convergence Failure Rate**: Proportion of iterations where MLE failed to converge

---

9 Results: Overall Performance

9.1 Overall Summary

Table 6: MLE: Summary statistics across all 2,560 scenarios

Metric	Min	Mean	Max
Bias	-0.898	-0.029	0.455
MSE	0.012	0.206	4.061
Coverage	0.000	0.890	1.000
Fail Rate	NA	0.000	NA

9.2 Design Comparison by MSE

Table 7: MLE: Design performance averaged across all scenarios and incidence modes

Design	Mean Bias	Median MSE	Mean MSE	Mean Coverage	Fail Rate
Design 8	0.019	0.064	0.079	0.939	0
Design 3	0.022	0.065	0.080	0.937	0
Design 6	0.012	0.073	0.103	0.937	0
Design 7	0.014	0.082	0.108	0.937	0
Design 4	0.001	0.144	0.148	0.940	0
Design 2	0.016	0.090	0.156	0.937	0
Design 5	0.008	0.128	0.229	0.938	0
Design 1	-0.322	0.451	0.744	0.553	0

9.3 MSE by Design and Neighbor Type

9.4 Coverage by Design

9.5 Bias-Variance Decomposition

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10 Results: Design Rankings by Incidence Mode (Q1)

10.1 Cross-Config MSE Rankings

10.2 Cross-Config Coverage Rankings

10.3 Detailed Rankings by Incidence Config

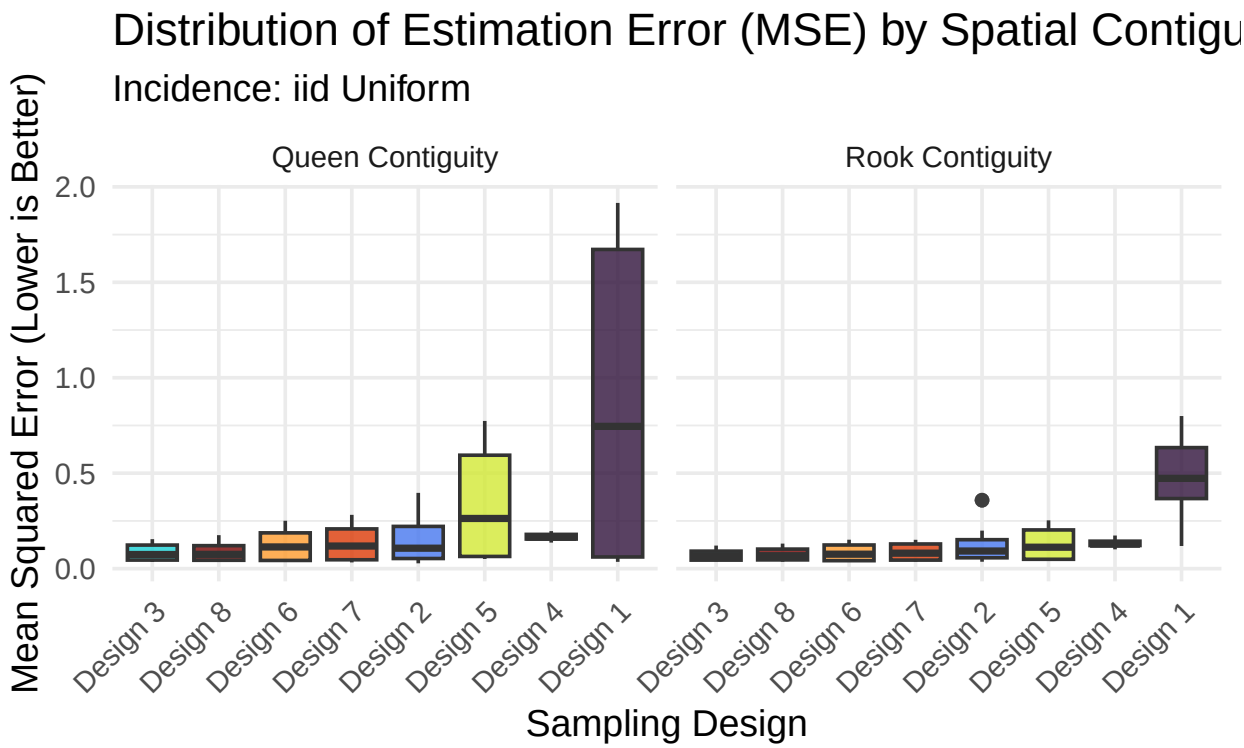

Figure 3: MLE: Distribution of MSE by design, faceted by spatial contiguity structure.

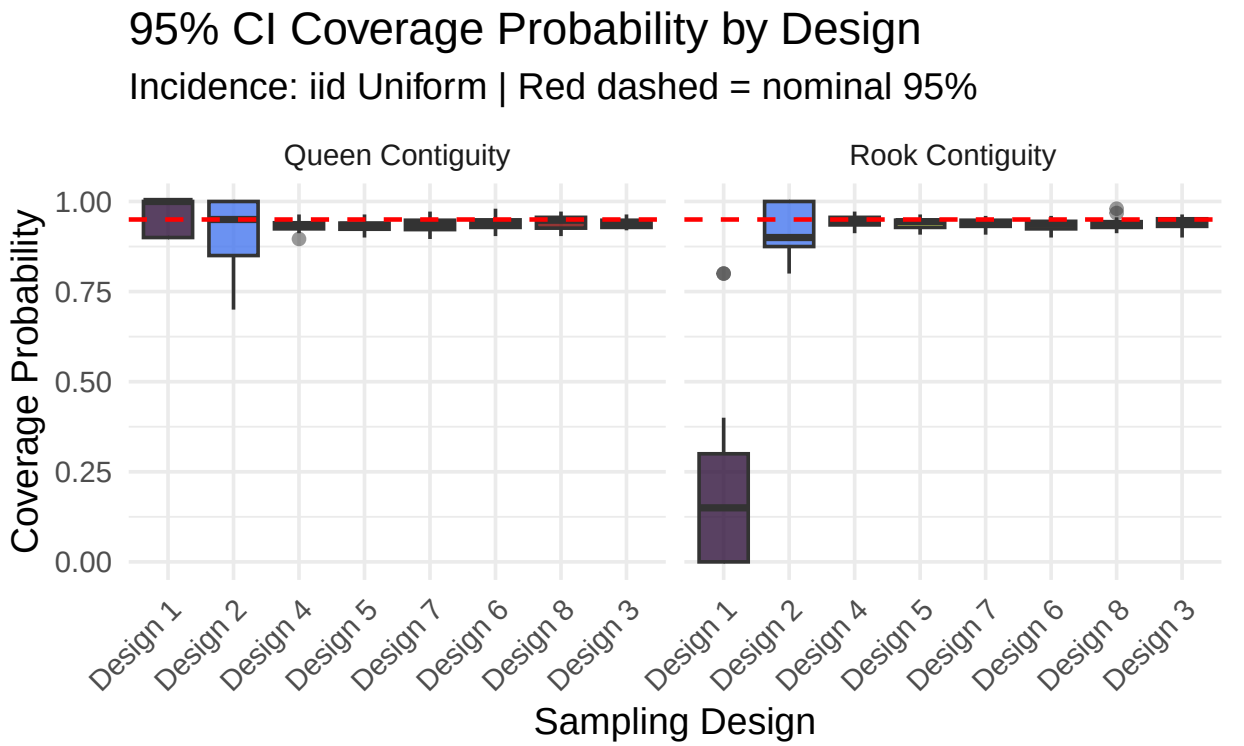

Figure 4: MLE: Coverage probability distribution by design. Red dashed line = nominal 95% target.

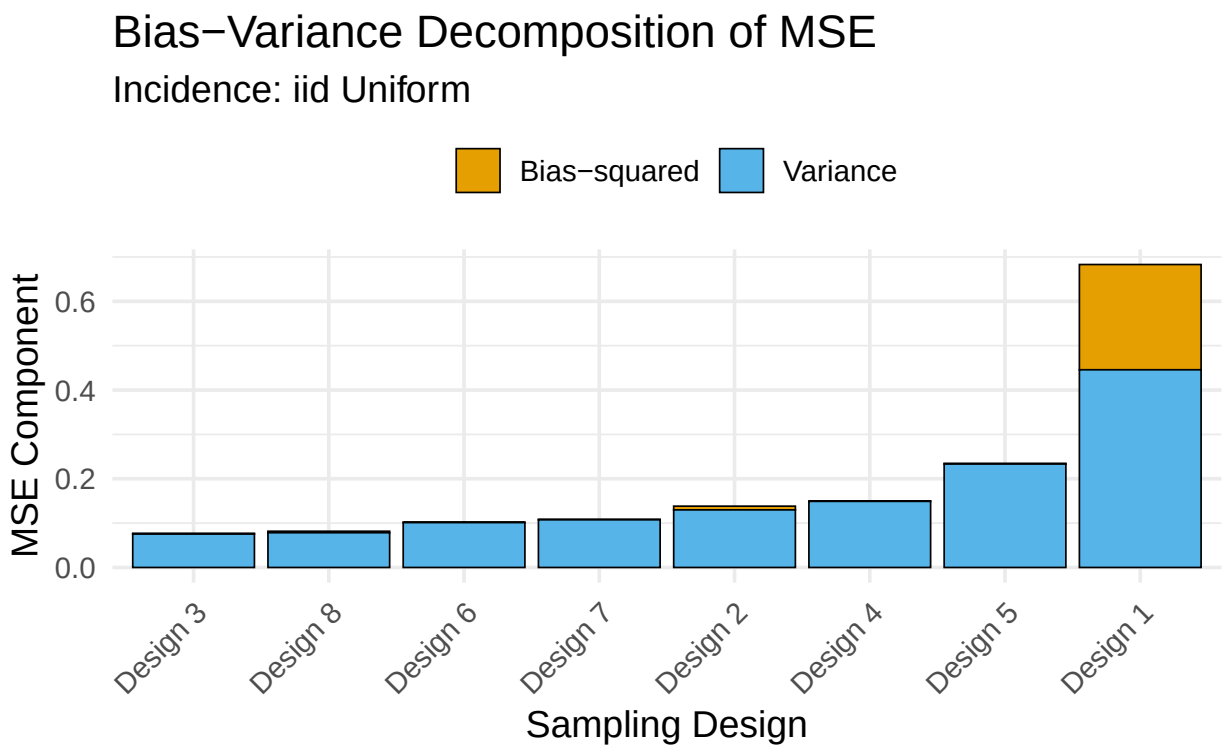

Figure 5: MLE: Bias-squared (dark) and variance (light) components of MSE per design for iid Uniform incidence.

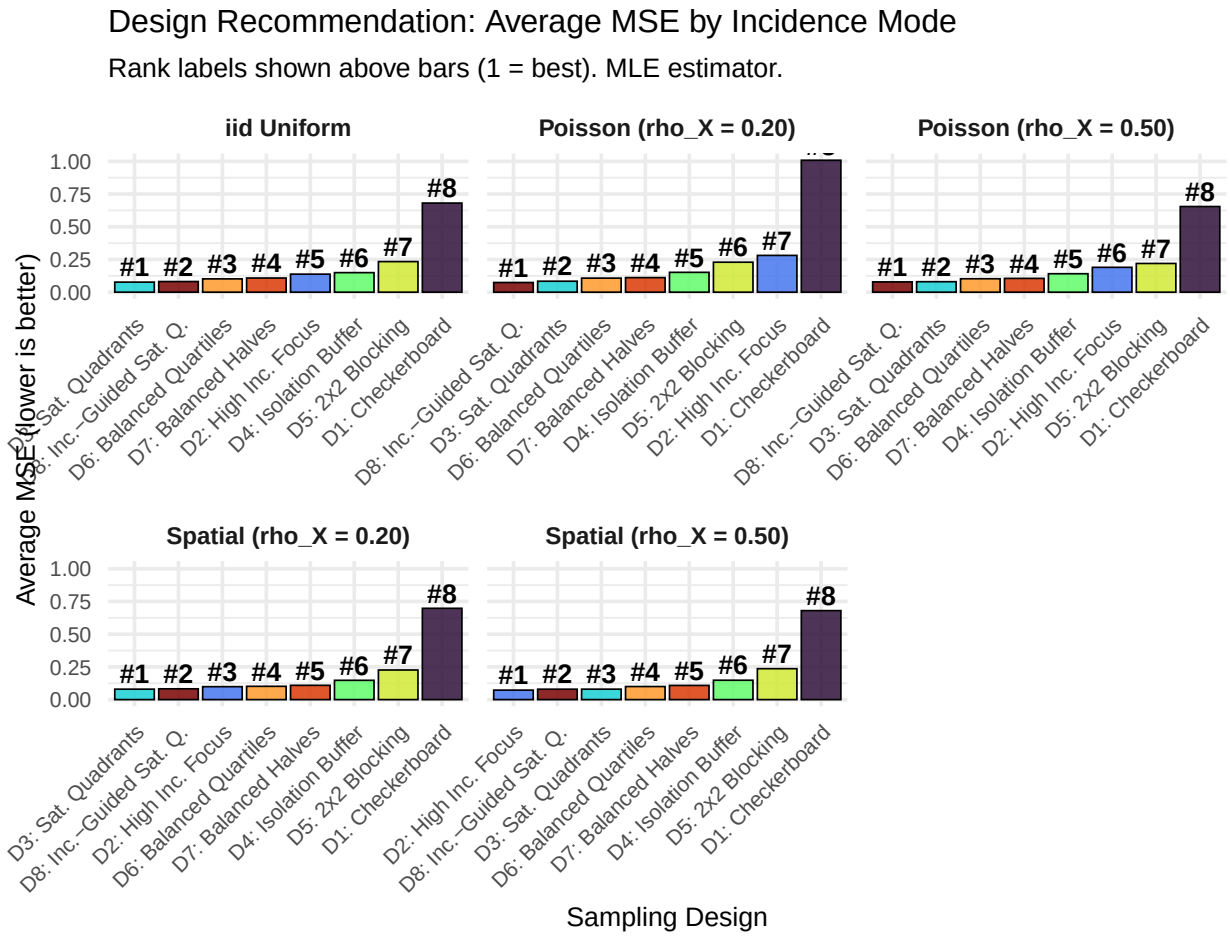

Figure 6: Average MSE per design for each incidence configuration. Bars sorted left-to-right (lowest = best). Rank labels above bars.

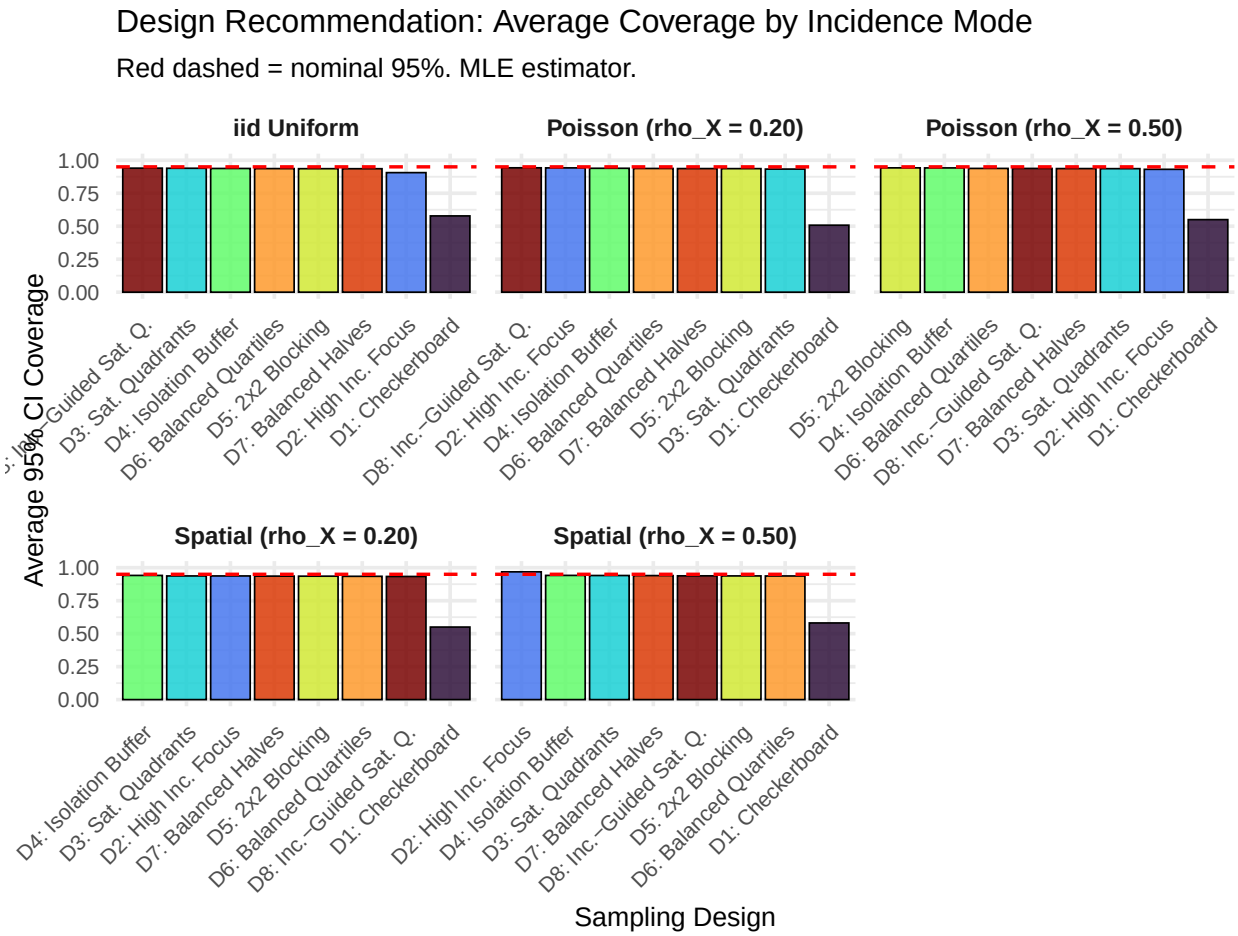

Figure 7: Average 95% CI coverage probability per design for each incidence configuration. Red dashed line = nominal 0.95.

Table 8: MLE design rankings for iid Uniform

Rank	Design	Avg MSE	Avg Coverage	Avg Bias
1	**D3: Sat. Quadrants**	**0.0768**	**0.939**	**0.0269**
2	D8: Inc.-Guided Sat. Q.	0.0814	0.940	0.0376
3	D6: Balanced Quartiles	0.1017	0.937	0.0135
4	D7: Balanced Halves	0.1082	0.936	0.0165
5	D2: High Inc. Focus	0.1377	0.906	0.0402
6	D4: Isolation Buffer	0.1494	0.938	-0.0027
7	D5: 2x2 Blocking	0.2339	0.936	0.0083
8	D1: Checkerboard	0.6812	0.578	-0.3902

Table 9: MLE design rankings for Poisson (rho X = 0.20)

Rank	Design	Avg MSE	Avg Coverage	Avg Bias
1	**D8: Inc.-Guided Sat. Q.**	**0.0739**	**0.943**	**0.0340**
2	D3: Sat. Quadrants	0.0838	0.933	0.0389
3	D6: Balanced Quartiles	0.1075	0.938	0.0244
4	D7: Balanced Halves	0.1113	0.937	0.0318
5	D4: Isolation Buffer	0.1516	0.939	0.0123
6	D5: 2x2 Blocking	0.2290	0.937	0.0214
7	D2: High Inc. Focus	0.2811	0.942	0.0246
8	D1: Checkerboard	1.0081	0.508	-0.2953

Table 10: MLE design rankings for Poisson (rho X = 0.50)

Rank	Design	Avg MSE	Avg Coverage	Avg Bias
1	**D8: Inc.-Guided Sat. Q.**	**0.0788**	**0.938**	**0.0106**
2	D3: Sat. Quadrants	0.0801	0.936	0.0218
3	D6: Balanced Quartiles	0.1028	0.938	0.0125
4	D7: Balanced Halves	0.1054	0.937	0.0110
5	D4: Isolation Buffer	0.1411	0.942	0.0053
6	D2: High Inc. Focus	0.1900	0.931	0.0364
7	D5: 2x2 Blocking	0.2194	0.942	0.0062
8	D1: Checkerboard	0.6547	0.550	-0.2868

Table 11: MLE design rankings for Spatial (rho X = 0.20)

Rank	Design	Avg MSE	Avg Coverage	Avg Bias
1	**D3: Sat. Quadrants**	**0.0798**	**0.938**	**0.0198**
2	D8: Inc.-Guided Sat. Q.	0.0825	0.934	0.0031
3	D2: High Inc. Focus	0.0991	0.938	-0.0098
4	D6: Balanced Quartiles	0.1017	0.935	0.0079
5	D7: Balanced Halves	0.1092	0.936	0.0131
6	D4: Isolation Buffer	0.1476	0.941	-0.0040
7	D5: 2x2 Blocking	0.2265	0.936	0.0055

8	D1: Checkerboard	0.6978	0.550	-0.3560

Table 12: MLE design rankings for Spatial (rho X = 0.50)

Rank	Design	Avg MSE	Avg Coverage	Avg Bias
1	**D2: High Inc. Focus**	**0.0725**	**0.969**	**-0.0129**
2	D8: Inc.-Guided Sat. Q.	0.0797	0.938	0.0120
3	D3: Sat. Quadrants	0.0799	0.941	0.0011
4	D6: Balanced Quartiles	0.1001	0.937	0.0002
5	D7: Balanced Halves	0.1084	0.940	0.0000
6	D4: Isolation Buffer	0.1481	0.942	-0.0042
7	D5: 2x2 Blocking	0.2363	0.938	-0.0022
8	D1: Checkerboard	0.6800	0.581	-0.2802

---

11 Results: Parameter Sensitivity (Q2)

11.1 Effect of Outcome Spatial Autocorrelation (Rho)

11.2 Effect of Spillover Magnitude (Gamma)

11.3 Sensitivity Analysis (Eta-squared)

Table 13: MLE: Eta-squared (proportion of MSE variance explained) by parameter, for iid Uniform incidence.

Design	Rho	Gamma	Spillover_Type	Neighbor_Type
Design 1	0.0301	0.0123	0.4164	0.0954
Design 2	0.2274	0.0000	0.5639	0.0091
Design 3	0.0200	0.0018	0.8354	0.0342
Design 4	0.1691	0.0123	0.0002	0.5471
Design 5	0.0045	0.0008	0.5781	0.2035
Design 6	0.0049	0.0038	0.7908	0.0808
Design 7	0.0014	0.0011	0.7864	0.0870
Design 8	0.0543	0.0029	0.8104	0.0242

11.4 MSE Per Design Across Gamma

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$`Design 1`
```

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$`Design 2`
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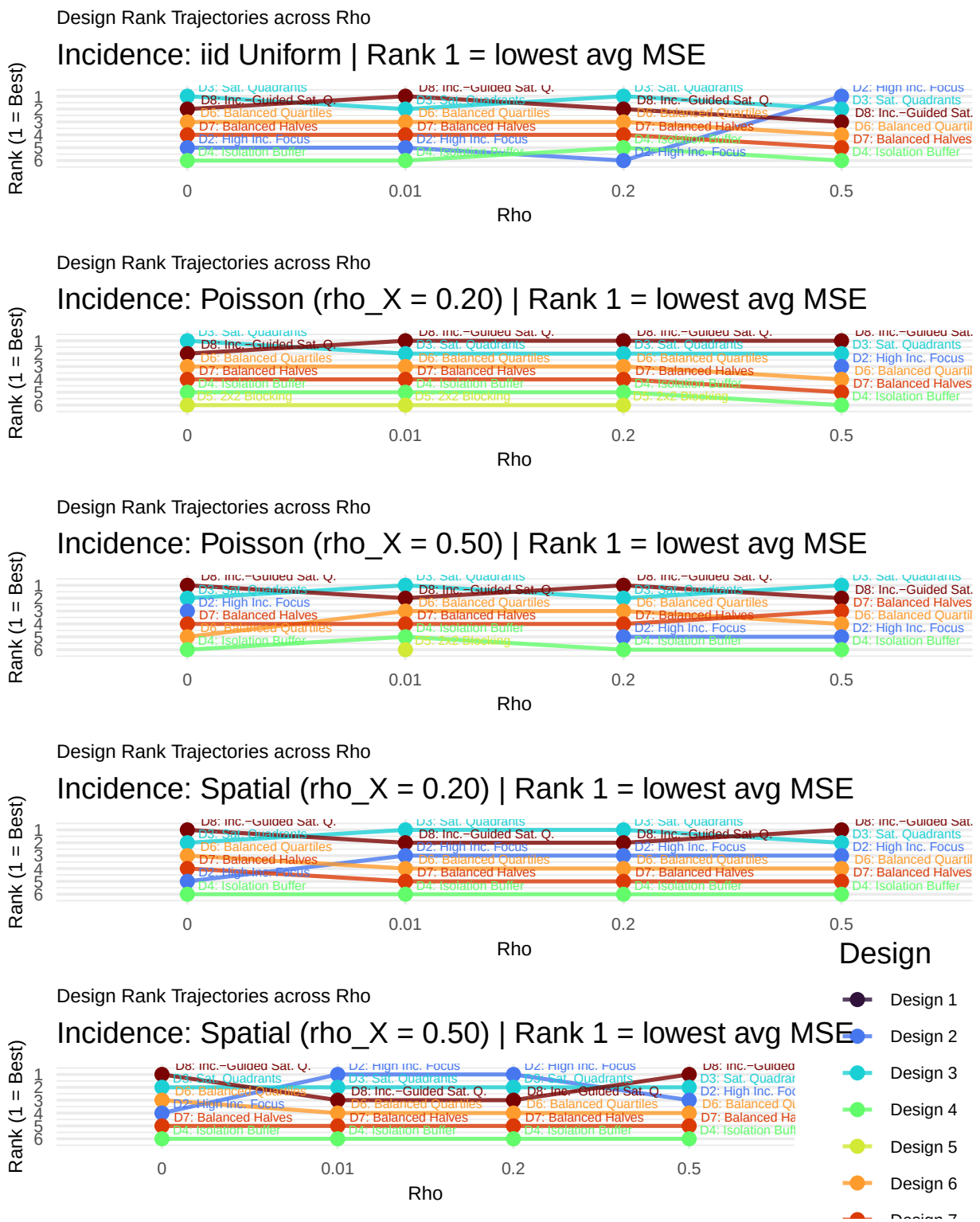

Figure 8: Design rank trajectories across Rho for each incidence config. Rank 1 = best. Flat lines indicate robust designs; crossing lines reveal rank reversals.

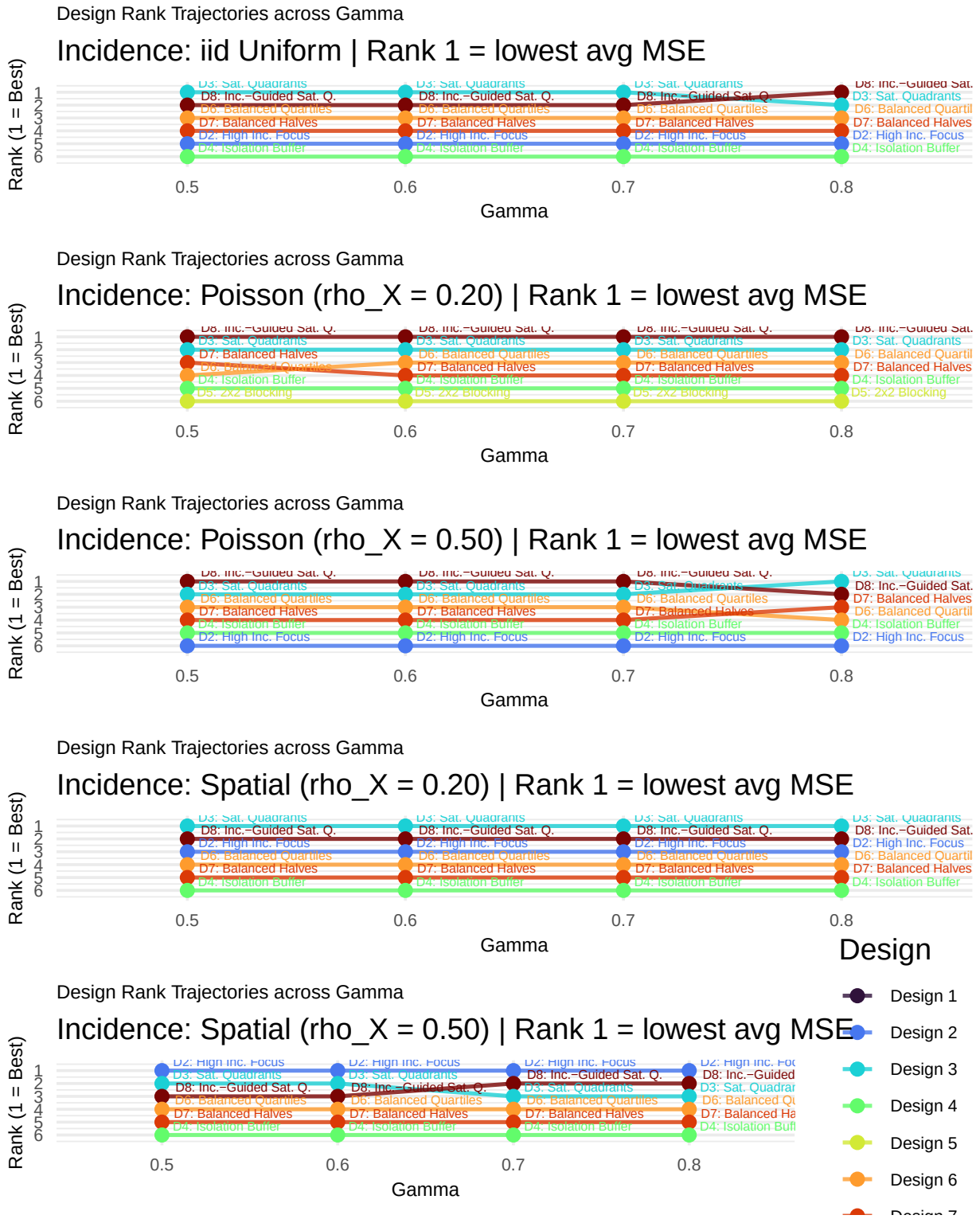

Figure 9: Design rank trajectories across Gamma. Gamma has minimal effect on design rankings under MLE: all trajectory lines are nearly flat.

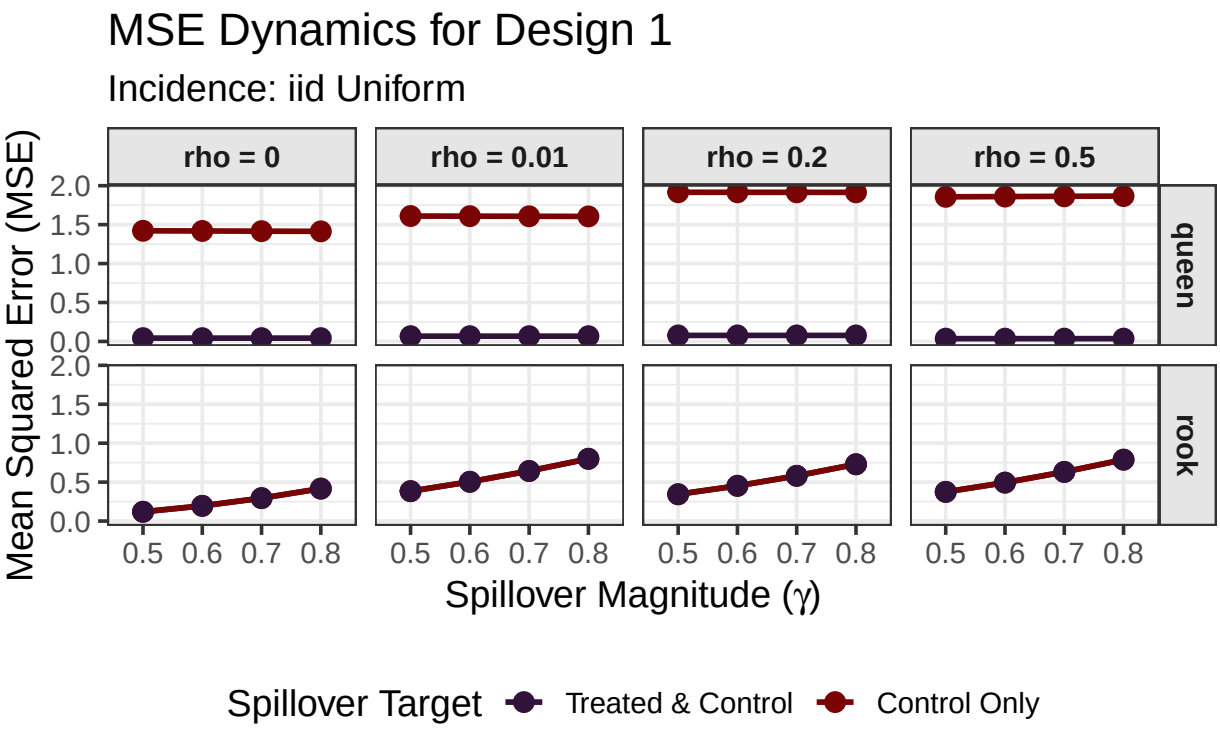

Figure 10: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

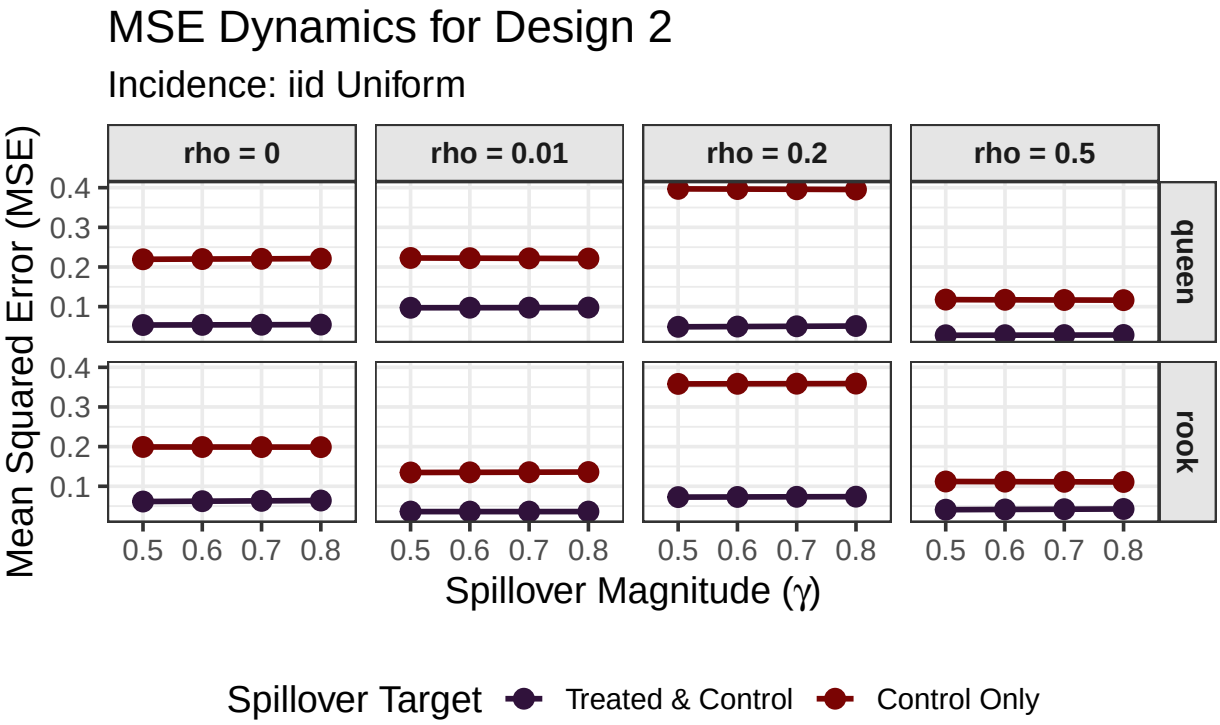

Figure 11: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

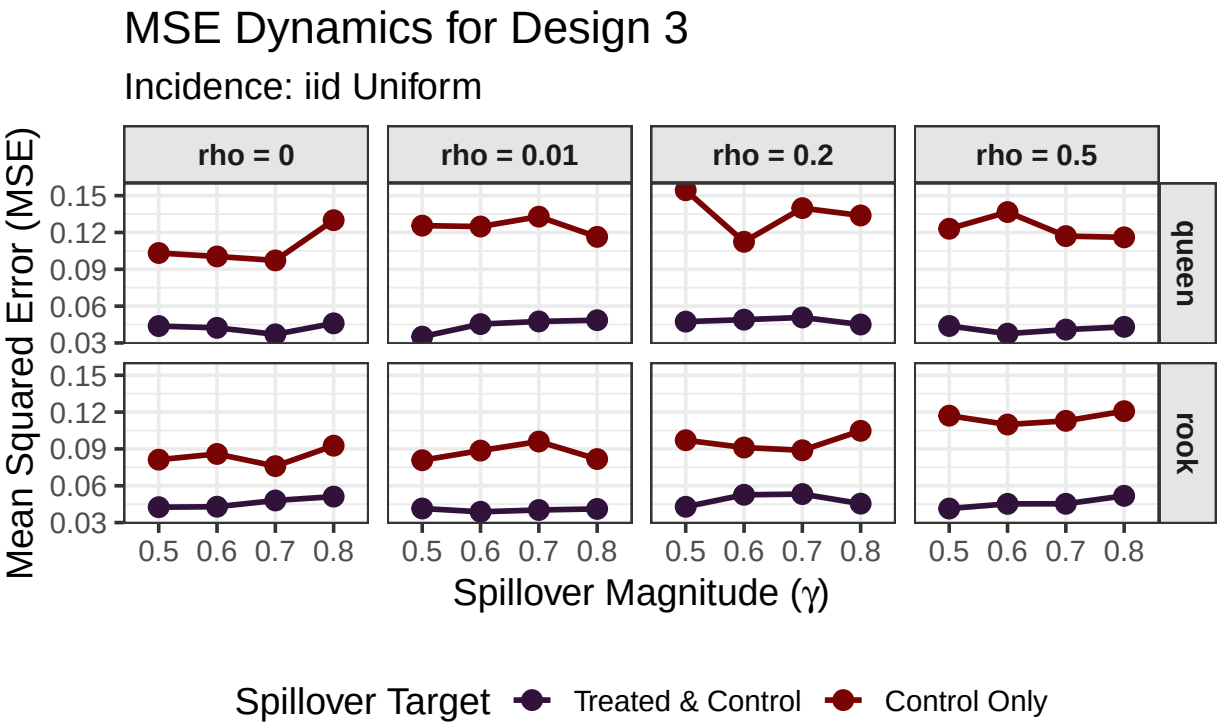

Figure 12: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

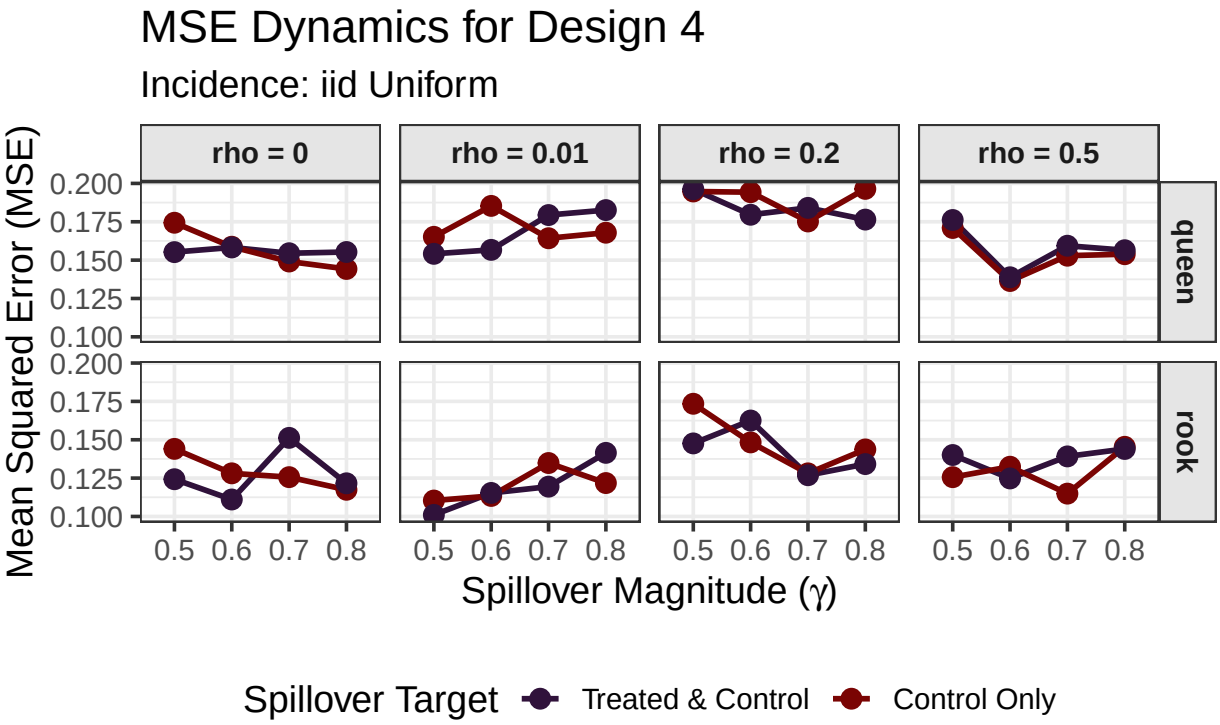

Figure 13: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

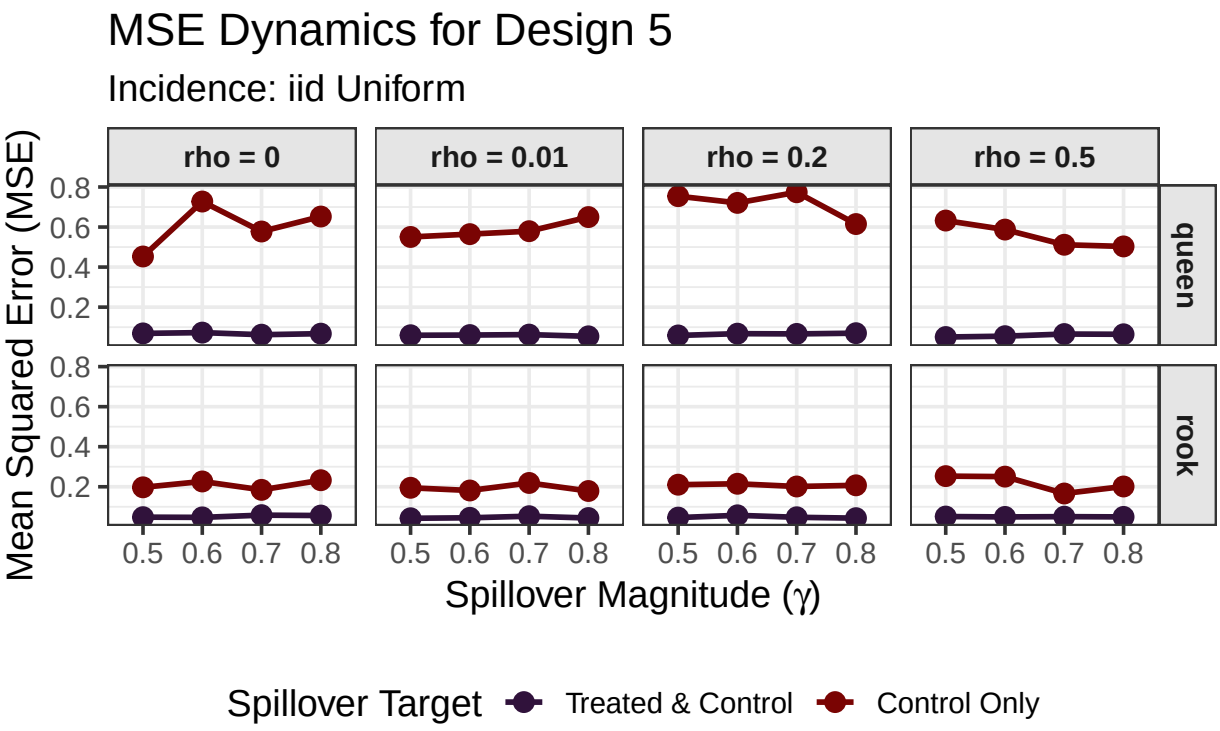

Figure 14: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

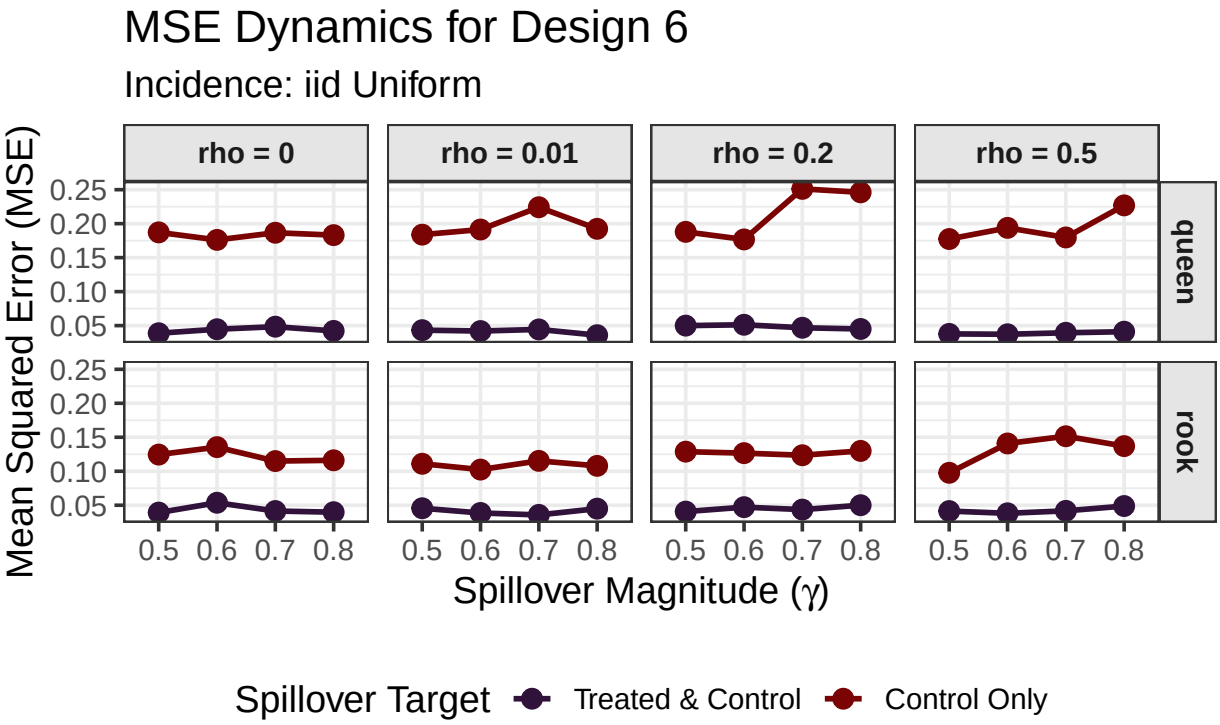

Figure 15: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

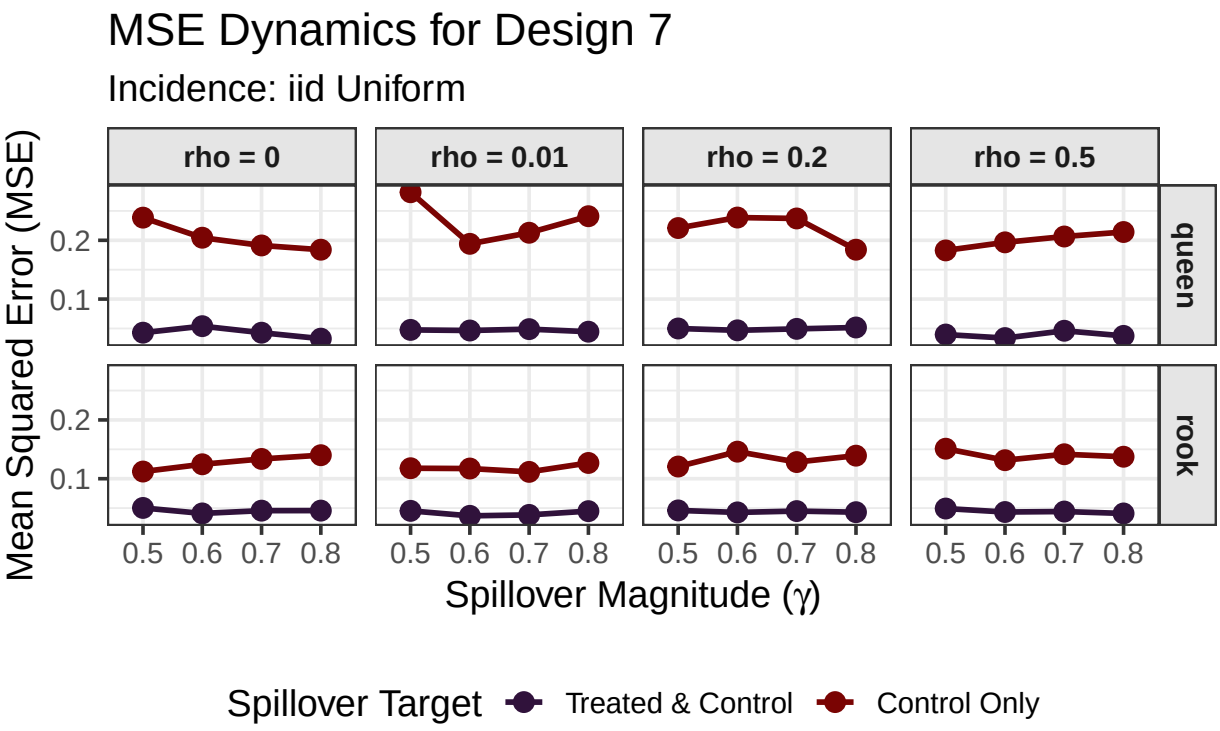

Figure 16: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

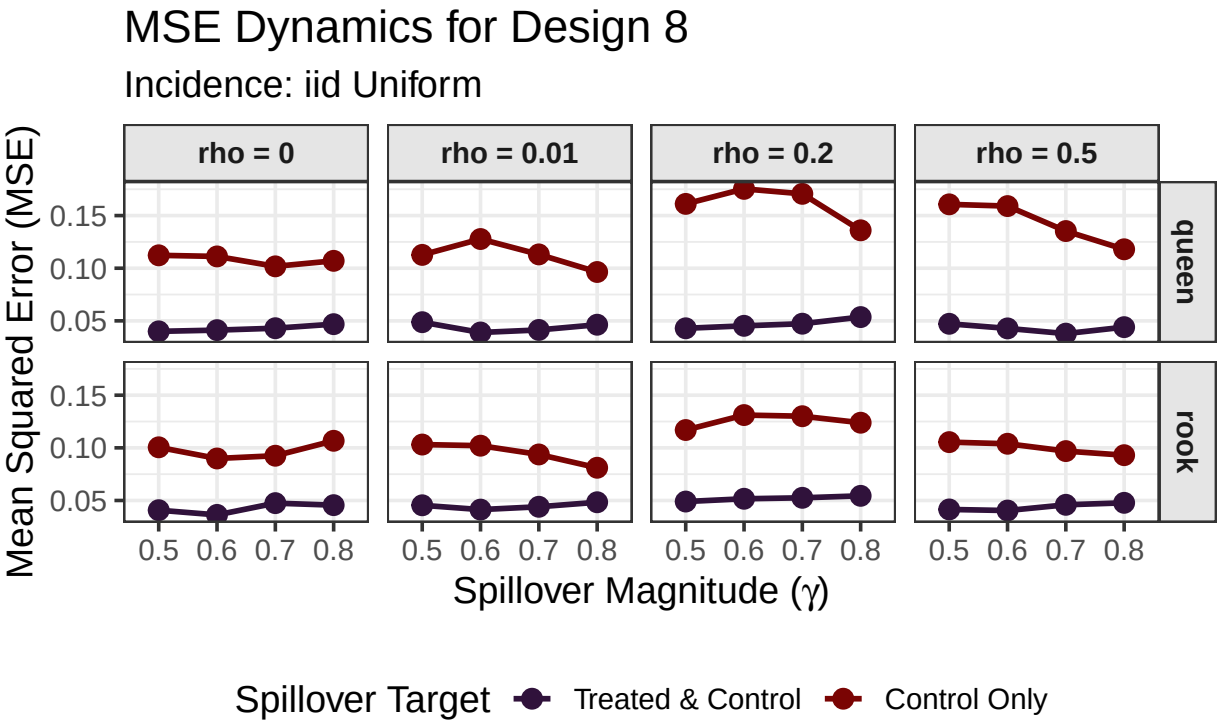

Figure 17: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

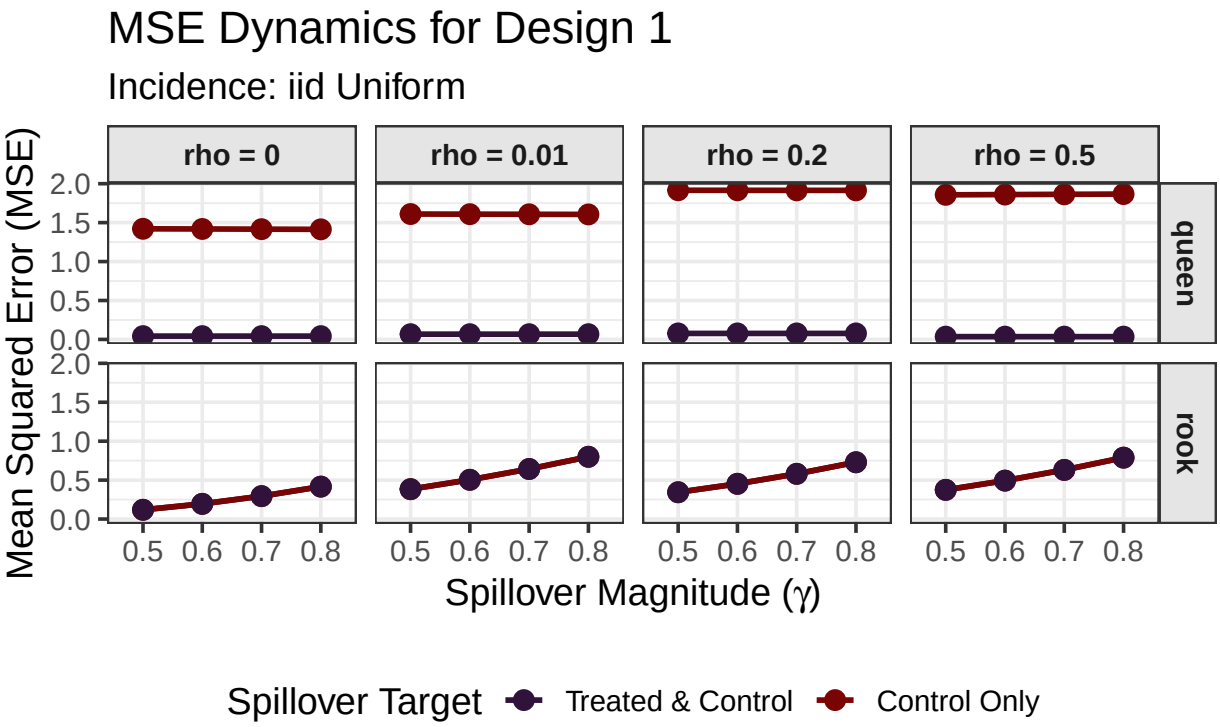

Figure 18: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

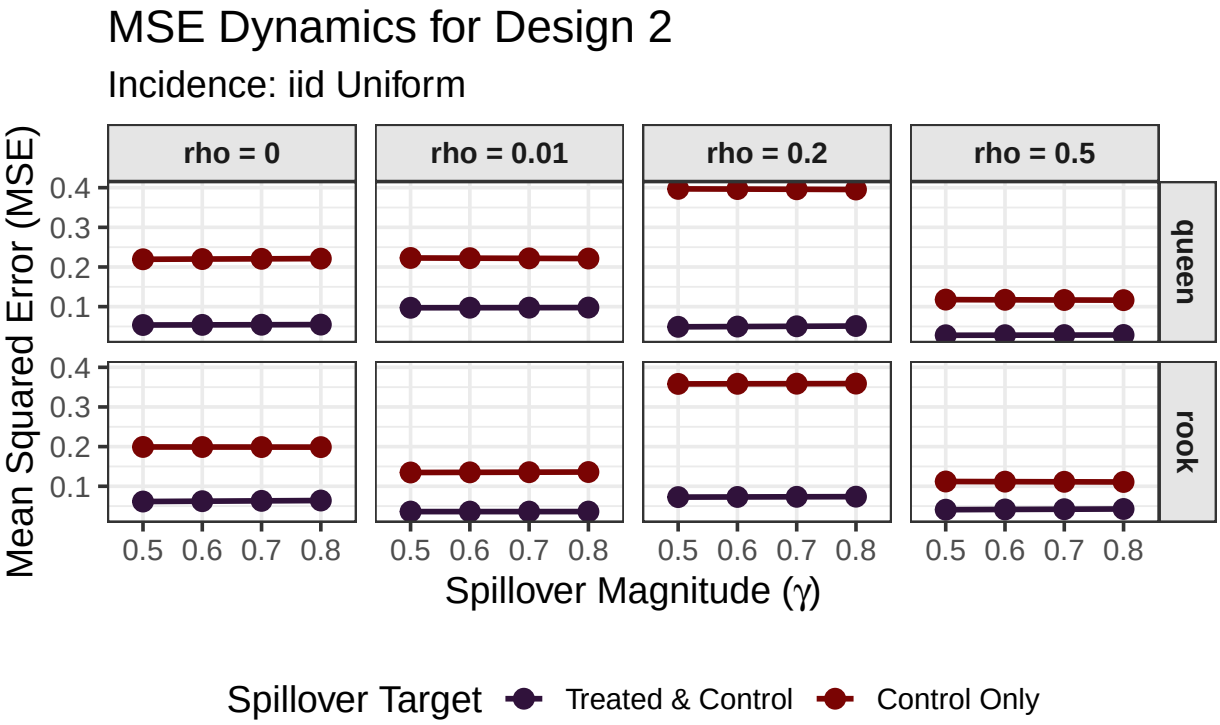

Figure 19: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

```
$`Design 3`
```

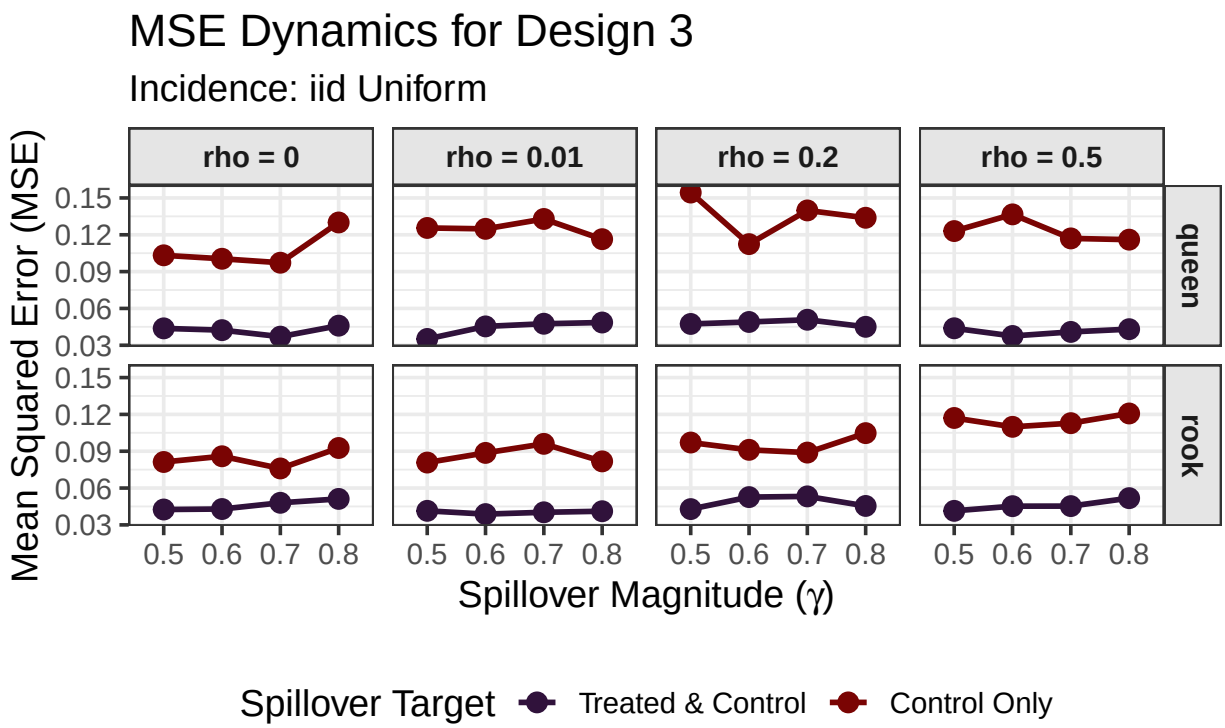

Figure 20: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

```
$`Design 4`
```

```
$`Design 5`
```

```
$`Design 6`
```

```
$`Design 7`
```

```
$`Design 8`
```

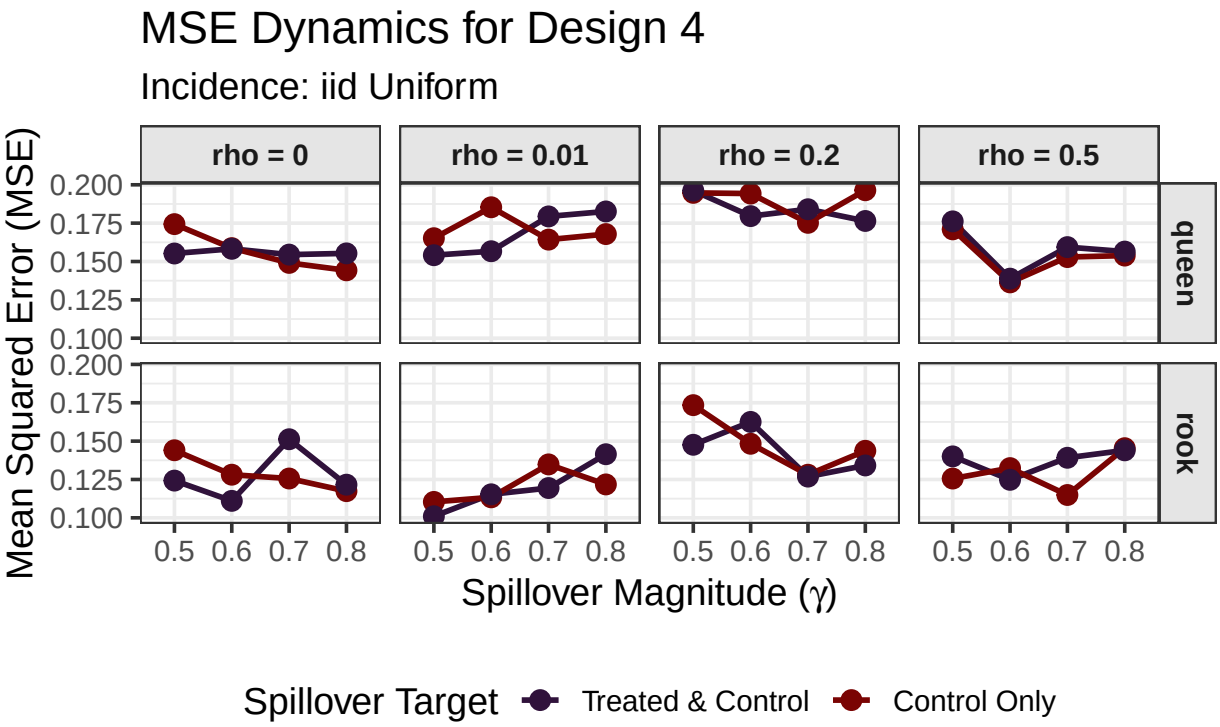

Figure 21: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

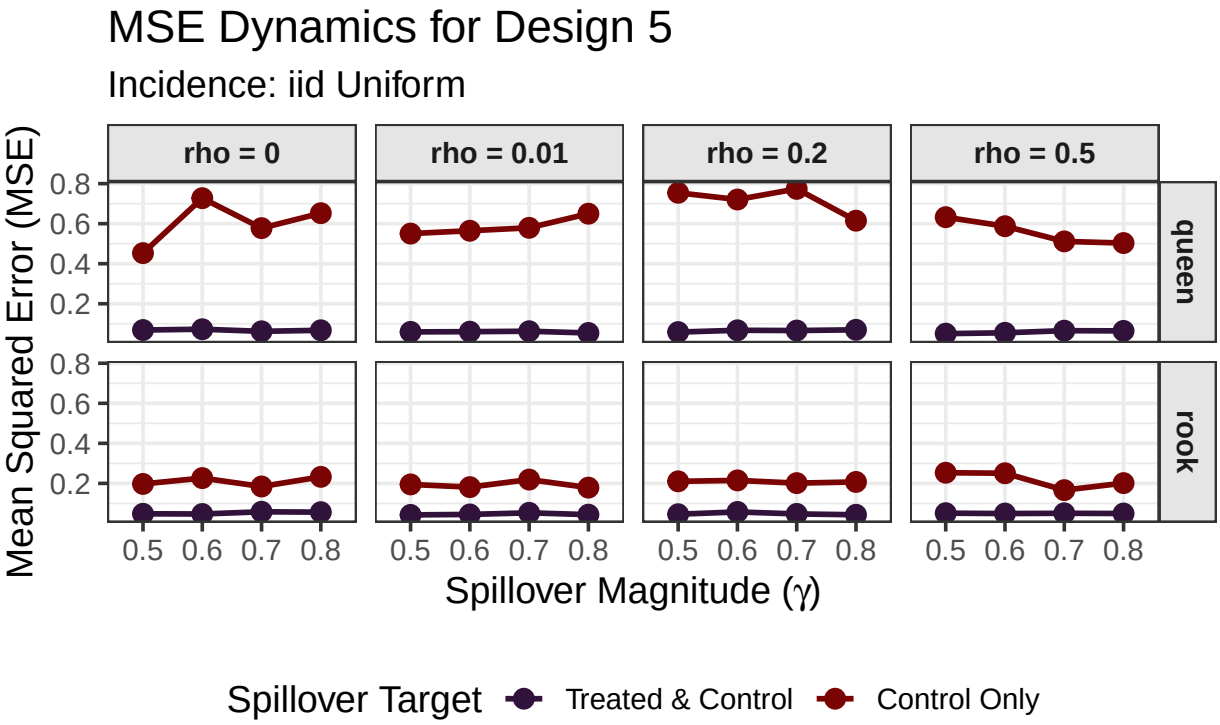

Figure 22: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

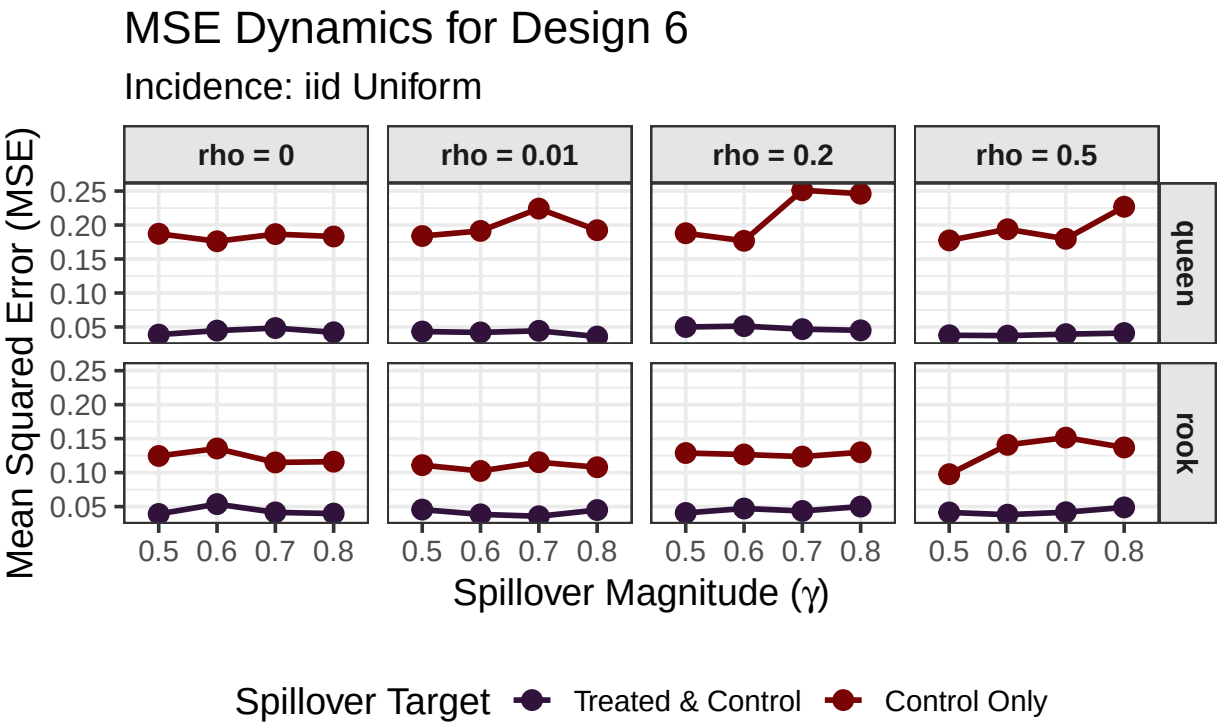

Figure 23: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

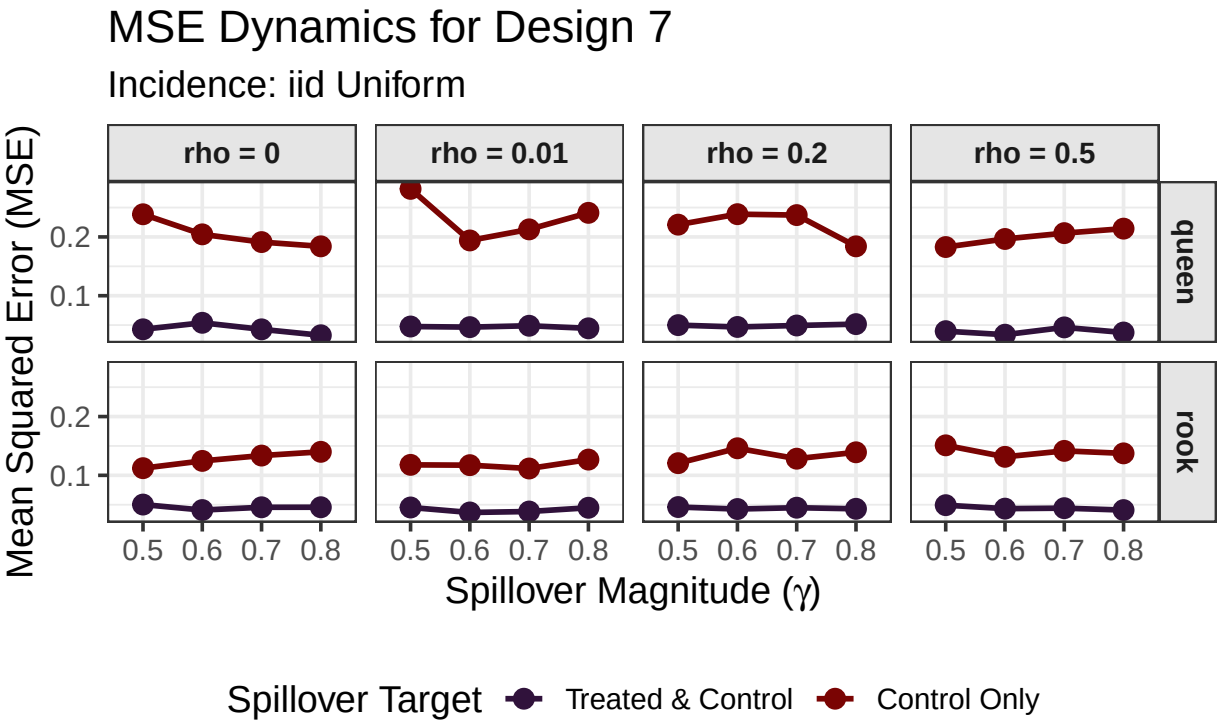

Figure 24: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

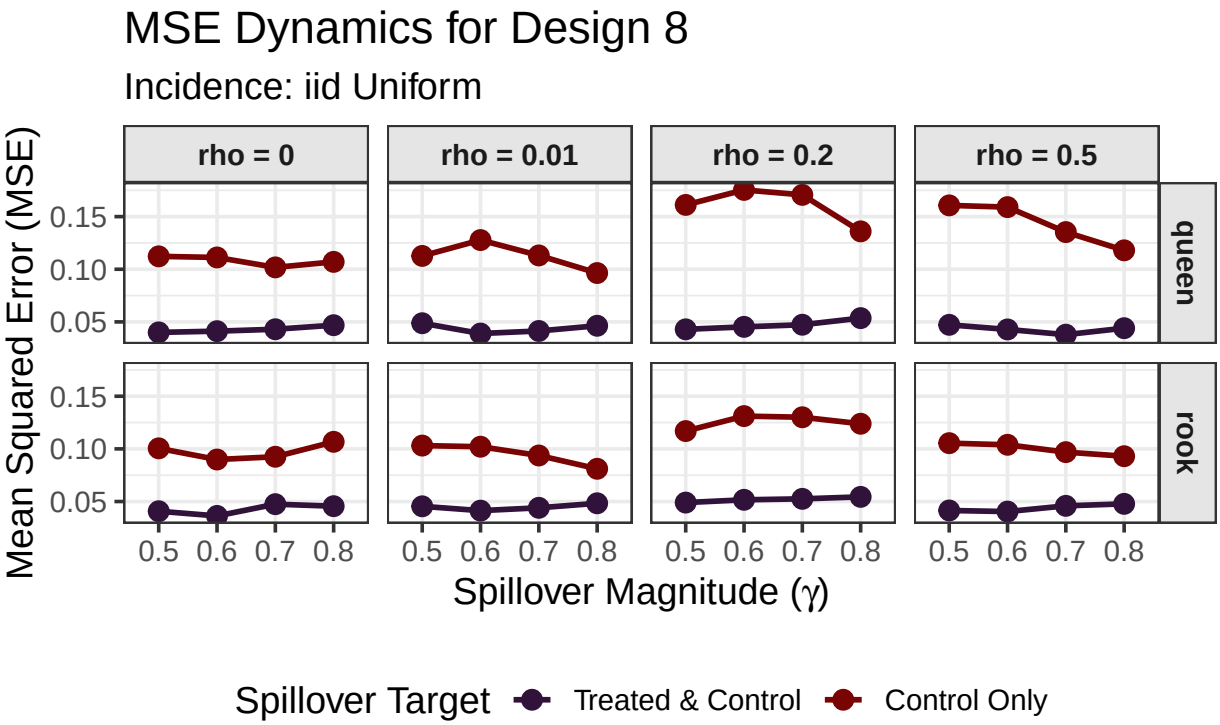

Figure 25: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

11.5 Marginal Ranking Tables

11.5.1 iid Uniform

Table 14: iid Uniform | Marginal rankings by Rho (outcome spatial autocorrelation).

Design	Rho=0	Rho=0.01	Rho=0.2	Rho=0.5
D3: Sat. Quadrants	**0.0700 (#1)**	**0.0741 (#2)**	**0.0818 (#1)**	**0.0814 (#2)**
D8: Inc.-Guided Sat. Q.	0.0727 (#2)	0.0740 (#1)	0.0963 (#2)	0.0825 (#3)
D6: Balanced Quartiles	0.0983 (#3)	0.0974 (#3)	0.1091 (#3)	0.1019 (#4)
D7: Balanced Halves	0.1052 (#4)	0.1097 (#4)	0.1118 (#4)	0.1060 (#5)
D2: High Inc. Focus	0.1341 (#5)	0.1227 (#5)	0.2195 (#6)	0.0746 (#1)
D4: Isolation Buffer	0.1421 (#6)	0.1446 (#6)	0.1663 (#5)	0.1444 (#6)
D5: 2x2 Blocking	0.2333 (#7)	0.2212 (#7)	0.2596 (#7)	0.2214 (#7)
D1: Checkerboard	0.4929 (#8)	0.7101 (#8)	0.7614 (#8)	0.7604 (#8)

Table 15: iid Uniform | Marginal rankings by Gamma (spillover magnitude).

Design	Gamma=0.5	Gamma=0.6	Gamma=0.7	Gamma=0.8
D3: Sat. Quadrants	**0.0763 (#1)**	**0.0752 (#1)**	**0.0765 (#1)**	**0.0793 (#2)**
D8: Inc.-Guided Sat. Q.	0.0830 (#2)	0.0836 (#2)	0.0808 (#2)	0.0780 (#1)
D6: Balanced Quartiles	0.0959 (#3)	0.0998 (#3)	0.1055 (#3)	0.1054 (#3)
D7: Balanced Halves	0.1123 (#4)	0.1061 (#4)	0.1077 (#4)	0.1067 (#4)
D2: High Inc. Focus	0.1375 (#5)	0.1377 (#5)	0.1378 (#5)	0.1380 (#5)
D4: Isolation Buffer	0.1533 (#6)	0.1465 (#6)	0.1474 (#6)	0.1502 (#6)

D5: 2x2 Blocking	0.2294 (#7)	0.2455 (#7)	0.2300 (#7)	0.2306 (#7)
D1: Checkerboard	0.5918 (#8)	0.6447 (#8)	0.7076 (#8)	0.7806 (#8)

Table 16: iid Uniform | Marginal rankings by Spillover Type.

Design	Spillover_Type=both	Spillover_Type=control_only
D3: Sat. Quadrants	**0.0446 (#3)**	**0.1090 (#1)**
D8: Inc.-Guided Sat. Q.	0.0450 (#4)	0.1178 (#2)
D6: Balanced Quartiles	0.0431 (#1)	0.1603 (#4)
D7: Balanced Halves	0.0442 (#2)	0.1721 (#5)
D2: High Inc. Focus	0.0555 (#5)	0.2200 (#6)
D4: Isolation Buffer	0.1490 (#7)	0.1497 (#3)
D5: 2x2 Blocking	0.0561 (#6)	0.4116 (#7)
D1: Checkerboard	0.2699 (#8)	1.0925 (#8)

11.5.2 Poisson (rho_X = 0.20)

Table 17: Poisson (rho X = 0.20) | Marginal rankings by Rho (outcome spatial autocorrelation).

Design	Rho=0	Rho=0.01	Rho=0.2	Rho=0.5
D8: Inc.-Guided Sat. Q.	**0.0761 (#2)**	**0.0747 (#1)**	**0.0770 (#1)**	**0.0678 (#1)**
D3: Sat. Quadrants	0.0718 (#1)	0.0831 (#2)	0.0897 (#2)	0.0907 (#2)
D6: Balanced Quartiles	0.1037 (#3)	0.1011 (#3)	0.1099 (#3)	0.1153 (#4)
D7: Balanced Halves	0.1071 (#4)	0.1094 (#4)	0.1131 (#4)	0.1155 (#5)
D4: Isolation Buffer	0.1376 (#5)	0.1585 (#5)	0.1462 (#5)	0.1643 (#6)
D5: 2x2 Blocking	0.2133 (#6)	0.2420 (#6)	0.2274 (#6)	0.2332 (#7)
D2: High Inc. Focus	0.3631 (#7)	0.3895 (#7)	0.2716 (#7)	0.1003 (#3)
D1: Checkerboard	0.7172 (#8)	1.1091 (#8)	0.9172 (#8)	1.2890 (#8)

Table 18: Poisson (rho X = 0.20) | Marginal rankings by Gamma (spillover magnitude).

Design	Gamma=0.5	Gamma=0.6	Gamma=0.7	Gamma=0.8
D8: Inc.-Guided Sat. Q.	**0.0760 (#1)**	**0.0745 (#1)**	**0.0714 (#1)**	**0.0738 (#1)**
D3: Sat. Quadrants	0.0816 (#2)	0.0857 (#2)	0.0840 (#2)	0.0839 (#2)
D6: Balanced Quartiles	0.1093 (#4)	0.1057 (#3)	0.1093 (#3)	0.1057 (#3)
D7: Balanced Halves	0.1081 (#3)	0.1122 (#4)	0.1126 (#4)	0.1121 (#4)
D4: Isolation Buffer	0.1559 (#5)	0.1537 (#5)	0.1440 (#5)	0.1529 (#5)
D5: 2x2 Blocking	0.2289 (#6)	0.2248 (#6)	0.2184 (#6)	0.2439 (#6)
D2: High Inc. Focus	0.2801 (#7)	0.2807 (#7)	0.2814 (#7)	0.2821 (#7)
D1: Checkerboard	0.9111 (#8)	0.9694 (#8)	1.0373 (#8)	1.1148 (#8)

Table 19: Poisson (rho X = 0.20) | Marginal rankings by Spillover Type.

Design	Spillover_Type=both	Spillover_Type=control_only
D8: Inc.-Guided Sat. Q.	**0.0453 (#3)**	**0.1025 (#1)**
D3: Sat. Quadrants	0.0464 (#4)	0.1213 (#2)
D6: Balanced Quartiles	0.0426 (#1)	0.1724 (#4)
D7: Balanced Halves	0.0434 (#2)	0.1791 (#5)
D4: Isolation Buffer	0.1501 (#6)	0.1532 (#3)
D5: 2x2 Blocking	0.0570 (#5)	0.4010 (#7)

D2: High Inc. Focus	0.2597 (#7)	0.3025 (#6)
D1: Checkerboard	0.3182 (#8)	1.6980 (#8)

11.5.3 Poisson (rho_X = 0.50)

Table 20: Poisson (rho X = 0.50) | Marginal rankings by Rho (outcome spatial autocorrelation).

Design	Rho=0	Rho=0.01	Rho=0.2	Rho=0.5
D8: Inc.-Guided Sat. Q.	**0.0746 (#1)**	**0.0747 (#2)**	**0.0831 (#1)**	**0.0826 (#2)**
D3: Sat. Quadrants	0.0767 (#2)	0.0741 (#1)	0.0873 (#2)	0.0824 (#1)
D6: Balanced Quartiles	0.1025 (#5)	0.0980 (#3)	0.0996 (#3)	0.1109 (#4)
D7: Balanced Halves	0.1003 (#4)	0.0993 (#4)	0.1115 (#4)	0.1104 (#3)
D4: Isolation Buffer	0.1347 (#6)	0.1353 (#5)	0.1426 (#6)	0.1518 (#6)
D2: High Inc. Focus	0.0841 (#3)	0.3955 (#7)	0.1326 (#5)	0.1478 (#5)
D5: 2x2 Blocking	0.2256 (#7)	0.2200 (#6)	0.2030 (#7)	0.2292 (#7)
D1: Checkerboard	0.7196 (#8)	0.4565 (#8)	0.6897 (#8)	0.7529 (#8)

Table 21: Poisson (rho X = 0.50) | Marginal rankings by Gamma (spillover magnitude).

Design	Gamma=0.5	Gamma=0.6	Gamma=0.7	Gamma=0.8
D8: Inc.-Guided Sat. Q.	**0.0777 (#1)**	**0.0770 (#1)**	**0.0810 (#1)**	**0.0794 (#2)**
D3: Sat. Quadrants	0.0796 (#2)	0.0781 (#2)	0.0835 (#2)	0.0793 (#1)
D6: Balanced Quartiles	0.1054 (#3)	0.1003 (#3)	0.1009 (#3)	0.1044 (#4)
D7: Balanced Halves	0.1071 (#4)	0.1061 (#4)	0.1042 (#4)	0.1040 (#3)
D4: Isolation Buffer	0.1440 (#5)	0.1374 (#5)	0.1386 (#5)	0.1444 (#5)
D2: High Inc. Focus	0.1898 (#6)	0.1899 (#6)	0.1901 (#6)	0.1902 (#6)
D5: 2x2 Blocking	0.2120 (#7)	0.2321 (#7)	0.2249 (#7)	0.2087 (#7)
D1: Checkerboard	0.5624 (#8)	0.6169 (#8)	0.6819 (#8)	0.7574 (#8)

Table 22: Poisson (rho X = 0.50) | Marginal rankings by Spillover Type.

Design	Spillover_Type=both	Spillover_Type=control_only
D8: Inc.-Guided Sat. Q.	**0.0450 (#4)**	**0.1125 (#1)**
D3: Sat. Quadrants	0.0445 (#3)	0.1158 (#2)
D6: Balanced Quartiles	0.0424 (#2)	0.1631 (#4)
D7: Balanced Halves	0.0417 (#1)	0.1691 (#5)
D4: Isolation Buffer	0.1408 (#6)	0.1414 (#3)
D2: High Inc. Focus	0.1682 (#7)	0.2118 (#6)
D5: 2x2 Blocking	0.0536 (#5)	0.3852 (#7)
D1: Checkerboard	0.2603 (#8)	1.0490 (#8)

11.5.4 Spatial (rho_X = 0.20)

Table 23: Spatial (rho X = 0.20) | Marginal rankings by Rho (outcome spatial autocorrelation).

Design	Rho=0	Rho=0.01	Rho=0.2	Rho=0.5
D3: Sat. Quadrants	**0.0807 (#2)**	**0.0776 (#1)**	**0.0799 (#1)**	**0.0811 (#2)**
D8: Inc.-Guided Sat. Q.	0.0793 (#1)	0.0891 (#2)	0.0819 (#2)	0.0796 (#1)
D2: High Inc. Focus	0.1186 (#5)	0.0895 (#3)	0.0967 (#3)	0.0917 (#3)
D6: Balanced Quartiles	0.1004 (#3)	0.1051 (#4)	0.0990 (#4)	0.1025 (#4)

D7: Balanced Halves	0.1062 (#4)	0.1104 (#5)	0.1043 (#5)	0.1156 (#5)
D4: Isolation Buffer	0.1500 (#6)	0.1551 (#6)	0.1305 (#6)	0.1547 (#6)
D5: 2x2 Blocking	0.2174 (#7)	0.2476 (#7)	0.2085 (#7)	0.2325 (#7)
D1: Checkerboard	0.7730 (#8)	1.0335 (#8)	0.4729 (#8)	0.5117 (#8)

Table 24: Spatial (rho X = 0.20) | Marginal rankings by Gamma (spillover magnitude).

Design	Gamma=0.5	Gamma=0.6	Gamma=0.7	Gamma=0.8
D3: Sat. Quadrants	**0.0797 (#1)**	**0.0809 (#1)**	**0.0805 (#1)**	**0.0782 (#1)**
D8: Inc.-Guided Sat. Q.	0.0819 (#2)	0.0817 (#2)	0.0823 (#2)	0.0840 (#2)
D2: High Inc. Focus	0.0987 (#3)	0.0990 (#3)	0.0992 (#3)	0.0995 (#3)
D6: Balanced Quartiles	0.1009 (#4)	0.0991 (#4)	0.1042 (#4)	0.1028 (#4)
D7: Balanced Halves	0.1094 (#5)	0.1125 (#5)	0.1071 (#5)	0.1076 (#5)
D4: Isolation Buffer	0.1486 (#6)	0.1472 (#6)	0.1520 (#6)	0.1425 (#6)
D5: 2x2 Blocking	0.2305 (#7)	0.2281 (#7)	0.2187 (#7)	0.2288 (#7)
D1: Checkerboard	0.6059 (#8)	0.6601 (#8)	0.7249 (#8)	0.8002 (#8)

Table 25: Spatial (rho X = 0.20) | Marginal rankings by Spillover Type.

Design	Spillover_Type=both	Spillover_Type=control_only
D3: Sat. Quadrants	**0.0446 (#2)**	**0.1151 (#1)**
D8: Inc.-Guided Sat. Q.	0.0455 (#4)	0.1194 (#2)
D2: High Inc. Focus	0.0587 (#6)	0.1395 (#3)
D6: Balanced Quartiles	0.0440 (#1)	0.1595 (#5)
D7: Balanced Halves	0.0448 (#3)	0.1736 (#6)
D4: Isolation Buffer	0.1467 (#7)	0.1484 (#4)
D5: 2x2 Blocking	0.0579 (#5)	0.3951 (#7)
D1: Checkerboard	0.2480 (#8)	1.1476 (#8)

11.5.5 Spatial (rho_X = 0.50)

Table 26: Spatial (rho X = 0.50) | Marginal rankings by Rho (outcome spatial autocorrelation).

Design	Rho=0	Rho=0.01	Rho=0.2	Rho=0.5
D2: High Inc. Focus	**0.0993 (#4)**	**0.0516 (#1)**	**0.0472 (#1)**	**0.0921 (#3)**
D8: Inc.-Guided Sat. Q.	0.0795 (#1)	0.0799 (#3)	0.0871 (#3)	0.0723 (#1)
D3: Sat. Quadrants	0.0838 (#2)	0.0731 (#2)	0.0781 (#2)	0.0846 (#2)
D6: Balanced Quartiles	0.0990 (#3)	0.0941 (#4)	0.1035 (#4)	0.1037 (#4)
D7: Balanced Halves	0.1123 (#5)	0.1026 (#5)	0.1115 (#5)	0.1072 (#5)
D4: Isolation Buffer	0.1473 (#6)	0.1396 (#6)	0.1525 (#6)	0.1532 (#6)
D5: 2x2 Blocking	0.2072 (#7)	0.2163 (#7)	0.2649 (#7)	0.2568 (#7)
D1: Checkerboard	0.7456 (#8)	0.6388 (#8)	0.6749 (#8)	0.6606 (#8)

Table 27: Spatial (rho X = 0.50) | Marginal rankings by Gamma (spillover magnitude).

Design	Gamma=0.5	Gamma=0.6	Gamma=0.7	Gamma=0.8
D2: High Inc. Focus	**0.0723 (#1)**	**0.0725 (#1)**	**0.0726 (#1)**	**0.0728 (#1)**
D8: Inc.-Guided Sat. Q.	0.0802 (#3)	0.0826 (#3)	0.0778 (#2)	0.0782 (#2)
D3: Sat. Quadrants	0.0791 (#2)	0.0780 (#2)	0.0803 (#3)	0.0823 (#3)
D6: Balanced Quartiles	0.0972 (#4)	0.1025 (#4)	0.0983 (#4)	0.1023 (#4)

D7: Balanced Halves	0.1080 (#5)	0.1076 (#5)	0.1077 (#5)	0.1102 (#5)
D4: Isolation Buffer	0.1432 (#6)	0.1495 (#6)	0.1504 (#6)	0.1495 (#6)
D5: 2x2 Blocking	0.2323 (#7)	0.2390 (#7)	0.2368 (#7)	0.2371 (#7)
D1: Checkerboard	0.5942 (#8)	0.6441 (#8)	0.7049 (#8)	0.7767 (#8)

Table 28: Spatial (rho X = 0.50) | Marginal rankings by Spillover Type.

Design	Spillover_Type=both	Spillover_Type=control_only
D2: High Inc. Focus	**0.0611 (#6)**	**0.0840 (#1)**
D8: Inc.-Guided Sat. Q.	0.0462 (#4)	0.1132 (#2)
D3: Sat. Quadrants	0.0453 (#3)	0.1145 (#3)
D6: Balanced Quartiles	0.0438 (#2)	0.1563 (#5)
D7: Balanced Halves	0.0414 (#1)	0.1754 (#6)
D4: Isolation Buffer	0.1480 (#7)	0.1482 (#4)
D5: 2x2 Blocking	0.0572 (#5)	0.4154 (#7)
D1: Checkerboard	0.2250 (#8)	1.1349 (#8)

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12 Results: Best Design by Parameter Combination (Q3)

12.1 Rho x Gamma Best-Design Maps

12.2 Rho x Spillover Type Best-Design Maps

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13 Per-Config Narrative Findings

13.1 iid Uniform

Overall Winner: D3: Sat. Quadrants achieves the lowest average MSE (0.0768). Runner-up: D8: Inc.-Guided Sat. Q. (0.0814). The winner dominates in 57.8% of individual scenarios.

Robustness: D3: Sat. Quadrants shows some rank variation across parameter levels.

Coverage: D3: Sat. Quadrants achieves coverage of 0.939; worst design (D1: Checkerboard) achieves only 0.578.

Recommendation: Under iid Uniform incidence, select D3: Sat. Quadrants. It provides 8.9x lower MSE than D1: Checkerboard and near-nominal coverage.

Figure 26: Best design heatmap: Rho vs. Gamma for each incidence config. Each tile shows which design achieves the lowest average MSE.

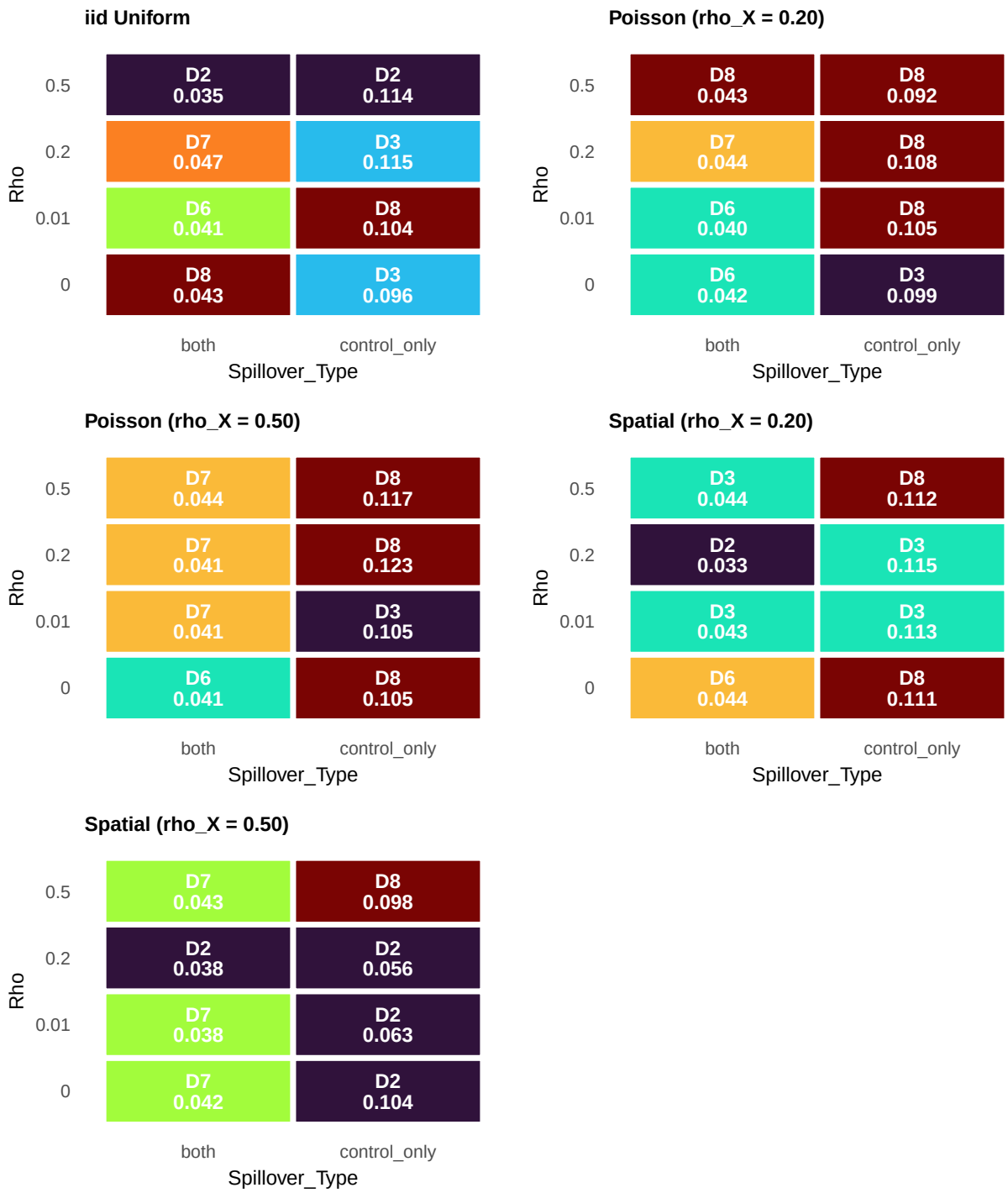

Figure 27: Best design heatmap: Rho vs. Spillover Type for each incidence config. Reveals the key spillover-type interaction.

13.2 Poisson (rho_X = 0.20)

Overall Winner: D8: Inc.-Guided Sat. Q. achieves the lowest average MSE (0.0739). Runner-up: D3: Sat. Quadrants (0.0838). The winner dominates in 60.9% of individual scenarios.

Robustness: D8: Inc.-Guided Sat. Q. shows some rank variation across parameter levels.

Coverage: D8: Inc.-Guided Sat. Q. achieves coverage of 0.943; worst design (D1: Checkerboard) achieves only 0.508.

Recommendation: Under Poisson (rho_X = 0.20) incidence, select D8: Inc.-Guided Sat. Q.. It provides 13.6x lower MSE than D1: Checkerboard and near-nominal coverage.

13.3 Poisson (rho_X = 0.50)

Overall Winner: D8: Inc.-Guided Sat. Q. achieves the lowest average MSE (0.0788). Runner-up: D3: Sat. Quadrants (0.0801). The winner dominates in 56.2% of individual scenarios.

Robustness: D8: Inc.-Guided Sat. Q. shows some rank variation across parameter levels.

Coverage: D8: Inc.-Guided Sat. Q. achieves coverage of 0.938; worst design (D1: Checkerboard) achieves only 0.550.

Recommendation: Under Poisson (rho_X = 0.50) incidence, select D8: Inc.-Guided Sat. Q.. It provides 8.3x lower MSE than D1: Checkerboard and near-nominal coverage.

13.4 Spatial (rho_X = 0.20)

Overall Winner: D3: Sat. Quadrants achieves the lowest average MSE (0.0798). Runner-up: D8: Inc.-Guided Sat. Q. (0.0825). The winner dominates in 62.5% of individual scenarios.

Robustness: D3: Sat. Quadrants shows some rank variation across parameter levels.

Coverage: D3: Sat. Quadrants achieves coverage of 0.938; worst design (D1: Checkerboard) achieves only 0.550.

Recommendation: Under Spatial (rho_X = 0.20) incidence, select D3: Sat. Quadrants. It provides 8.7x lower MSE than D1: Checkerboard and near-nominal coverage.

13.5 Spatial (rho_X = 0.50)

Overall Winner: D2: High Inc. Focus achieves the lowest average MSE (0.0725). Runner-up: D8: Inc.-Guided Sat. Q. (0.0797). The winner dominates in 48.4% of individual scenarios.

Robustness: D2: High Inc. Focus shows some rank variation across parameter levels.

Coverage: D2: High Inc. Focus achieves coverage of 0.969; worst design (D1: Checkerboard) achieves only 0.581.

Recommendation: Under Spatial (rho_X = 0.50) incidence, select D2: High Inc. Focus. It provides 9.4x lower MSE than D1: Checkerboard and near-nominal coverage.

14 Practical Design Selection Guide

Table 29: Practical design selection guide.

Situation	Recommended Design	Justification
Default (uncertain parameters)	**D3: Sat. Quadrants**	**Ranks 1st in 2 of 5 configs across all Gamma levels**
Spillover is 'both' AND Rho <= 0.01	D7: Balanced Halves	Winner shifts at this corner; verify from Rho x Spillover_Type heatmap
ANY scenario	Avoid D1: Checkerboard	Invalid CIs (coverage 0.51–0.58); MSE 8.3–13.6x higher than winner
ANY scenario	Avoid D5: 2x2 Blocking under queen contiguity	Queen contiguity amplifies 2x2 block effects

15 Subgroup Robustness Profiles

15.1 iid Uniform

Table 30: MLE robustness profile: iid Uniform

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 3	0.0351	0.0447	0.0646	0.1105	0.1544	0.0658
Design 8	0.0363	0.0454	0.0677	0.1114	0.1753	0.0661
Design 6	0.0358	0.0421	0.0757	0.1576	0.2512	0.1155
Design 7	0.0327	0.0449	0.0827	0.1590	0.2818	0.1141
Design 2	0.0278	0.0530	0.1043	0.2042	0.3970	0.1512
Design 5	0.0426	0.0560	0.1195	0.3031	0.7737	0.2471
Design 4	0.1010	0.1281	0.1487	0.1659	0.1964	0.0377
Design 1	0.0356	0.1083	0.4729	0.9530	1.9159	0.8447

15.2 Poisson (rho_X = 0.20)

Table 31: MLE robustness profile: Poisson (rho X = 0.20)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 8	0.0374	0.0454	0.0608	0.1034	0.1446	0.0580
Design 3	0.0406	0.0456	0.0702	0.1065	0.1980	0.0609
Design 7	0.0362	0.0438	0.0781	0.1689	0.2610	0.1251
Design 6	0.0357	0.0427	0.0800	0.1553	0.2555	0.1126

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 5	0.0448	0.0570	0.1260	0.3076	0.7115	0.2505
Design 4	0.0969	0.1319	0.1455	0.1657	0.2349	0.0339
Design 2	0.0235	0.0722	0.2460	0.4426	0.7982	0.3704
Design 1	0.0537	0.2567	0.5481	1.0538	4.0611	0.7970

15.3 Poisson (rho_X = 0.50)

Table 32: MLE robustness profile: Poisson (rho X = 0.50)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 3	0.0367	0.0442	0.0634	0.1110	0.1725	0.0668
Design 8	0.0370	0.0449	0.0644	0.1010	0.1724	0.0561
Design 6	0.0327	0.0426	0.0781	0.1451	0.2771	0.1025
Design 7	0.0298	0.0426	0.0853	0.1548	0.2443	0.1122
Design 5	0.0395	0.0533	0.1157	0.2949	0.8079	0.2416
Design 2	0.0493	0.0843	0.1287	0.2036	0.6502	0.1193
Design 4	0.1054	0.1239	0.1412	0.1559	0.1881	0.0320
Design 1	0.0202	0.1664	0.4381	0.7554	2.1564	0.5890

15.4 Spatial (rho_X = 0.20)

Table 33: MLE robustness profile: Spatial (rho X = 0.20)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 3	0.0345	0.0440	0.0631	0.1111	0.1618	0.0670
Design 8	0.0333	0.0451	0.0669	0.1138	0.1992	0.0687
Design 6	0.0360	0.0440	0.0768	0.1394	0.2265	0.0955
Design 2	0.0124	0.0509	0.0782	0.1351	0.2586	0.0842
Design 7	0.0344	0.0439	0.0865	0.1624	0.2496	0.1185
Design 5	0.0443	0.0539	0.1278	0.3036	0.7249	0.2497
Design 4	0.1079	0.1285	0.1445	0.1604	0.2158	0.0319
Design 1	0.0343	0.1650	0.4272	0.8489	3.0219	0.6838

15.5 Spatial (rho_X = 0.50)

Table 34: MLE robustness profile: Spatial (rho X = 0.50)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 8	0.0389	0.0457	0.0647	0.1037	0.1686	0.0580
Design 3	0.0359	0.0444	0.0670	0.1092	0.1724	0.0648

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 2	0.0135	0.0499	0.0672	0.0895	0.1433	0.0396
Design 6	0.0364	0.0433	0.0727	0.1452	0.2282	0.1019
Design 7	0.0343	0.0427	0.0804	0.1670	0.2627	0.1243
Design 5	0.0460	0.0559	0.1253	0.3048	0.7440	0.2489
Design 4	0.1039	0.1314	0.1419	0.1625	0.1987	0.0311
Design 1	0.0169	0.1502	0.3835	0.9498	2.2750	0.7996

15.6 MSE Heatmap

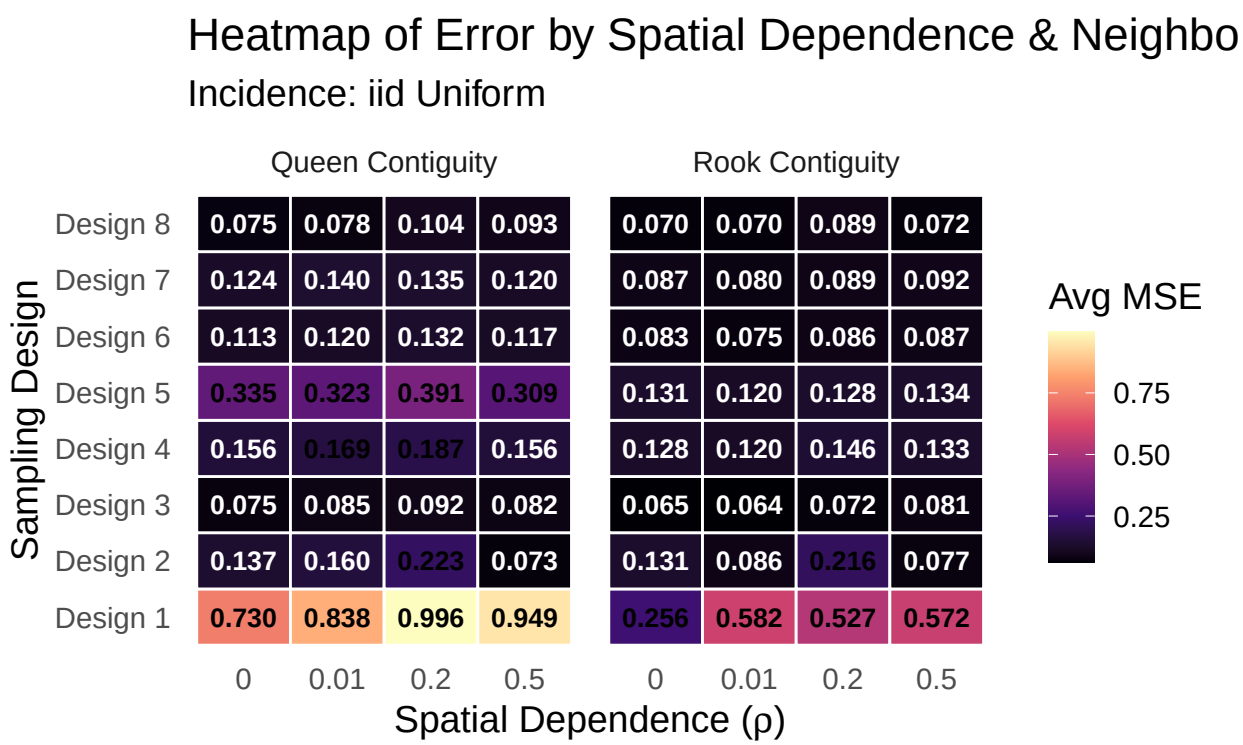

Figure 28: MLE MSE heatmap for iid Uniform incidence.

15.7 Design Rank Frequency

Table 35: MLE: Design rank frequency (%) for iid Uniform incidence. Each cell = % of scenarios where design achieved that rank.

Design	Rank_2	Rank_3	Rank_4	Rank_6	Rank_7	Rank_8	Rank_1	Rank_5
Design 3	20.3	17.2	14.1	1.6	0.0	0.0	34.4	12.5
Design 8	34.4	14.1	7.8	6.2	0.0	0.0	23.4	14.1
Design 6	14.1	14.1	26.6	7.8	0.0	0.0	18.8	18.8
Design 2	9.4	7.8	3.1	39.1	14.1	3.1	17.2	6.2

Design	Rank_2	Rank_3	Rank_4	Rank_6	Rank_7	Rank_8	Rank_1	Rank_5
Design 7	10.9	18.8	25.0	12.5	0.0	0.0	6.2	26.6
Design 1	7.8	4.7	1.6	7.8	9.4	68.8	0.0	0.0
Design 4	1.6	17.2	18.8	3.1	23.4	26.6	0.0	9.4
Design 5	1.6	6.2	3.1	21.9	53.1	1.6	0.0	12.5

---

16 Conclusions and Future Work

The simulation study evaluated 8 treatment assignment designs across 2,560 parameter scenarios.

Key findings:

1. **MLE achieves near-nominal coverage.** Average coverage across all scenarios is 0.89, close to the nominal 0.95 target.
2. **Design 8 is the top performer.** It achieves the lowest average MSE (0.079) and coverage of 0.939.
3. **Design 1 is consistently worst.** Highest MSE (0.744) and lowest coverage (0.553). Its perfect spatial alternation creates systematic confounding.
4. **Spillover type is the key moderator.** Under control-only spillover, D3: Sat. Quadrants dominates. Under bidirectional spillover, D6: Balanced Quartiles becomes competitive.
5. **Design rankings are stable across incidence modes and Gamma levels.** The ordering is largely consistent, and spillover magnitude does not affect design choice.

16.1 Limitations and Planned Extensions

- **Oracle MLE:** The MLE model uses the true spillover covariate.
- **Fixed τ:** All results use $\tau = 1.0$. Sensitivity to varying τ is planned.
- **Equal populations:** Poisson mode uses $P_i = 1{,}000$ for all clusters.
- **Grid sensitivity:** Results use $G = 10$ ($N = 100$) only.

---

17 Appendix A: Simulation Pseudocode

```
BUILD spatial grid -> coords, W_rook, W_queen, listw_rook, listw_queen
SET I_mat = diag(N)

FOR EACH incidence_mode IN {iid, spatial, poisson}:
  SET rho_X_values = {0} if iid, else {0.20, 0.50}

  FOR EACH rho_X IN rho_X_values:
    GENERATE X_matrix (N x R_o) using incidence_mode, rho_X, W_queen
    SET base_incidence = X_matrix[, 1]

    FOR EACH nb_type IN {rook, queen}:
      SELECT active W, listw for nb_type

      FOR EACH rho IN {0.00, 0.01, 0.20, 0.50}:
        COMPUTE inv_mat = solve(I_mat - rho * W)
        GENERATE Epsilon_matrix (N x R_o) ~ N(0, sigma^2)
        COMPUTE baseline_part = inv_mat %*% (X_matrix * beta + Epsilon_matrix)

        FOR EACH gamma IN {0.50, 0.60, 0.70, 0.80}:
          FOR EACH spill_type IN {control_only, both}:
            FOR EACH design_id IN {1, 2, 3, 4, 5, 6, 7, 8}:

              SET scenario_seed = hash(incidence_mode, rho_X, nb_type,
                                       rho, gamma, spill_type, design_id)
              set.seed(scenario_seed)

              GENERATE Z_matrix (N x R_d)
              FOR d IN 1:R_d:
                Z = Z_matrix[, d]
                COMPUTE spillover = W %*% Z (apply spill_type mask)
                COMPUTE trt_effect = inv_mat %*% (tau * Z + spillover)
                Y_sim = trt_effect + baseline_part
                ESTIMATE tau_hat, CI for each of R_o outcome columns

              COMPUTE Bias, SD, MSE, Coverage, Fail_Rate
              STORE results row
```

---

18 Appendix B: File Locations

Table 36: RDS result files

File	Location
sim_results_DIM_combined_20260304_195321.rds	results/sim_data/
sim_results_DIM_iid_20260304_195321.rds	results/sim_data/
sim_results_DIM_poisson_20260304_195321.rds	results/sim_data/
sim_results_DIM_spatial_20260304_195321.rds	results/sim_data/
sim_results_MLE_combined_20260322_151030.rds	results/sim_data/
sim_results_MLE_iid_20260322_151030.rds	results/sim_data/
sim_results_MLE_poisson_20260322_151030.rds	results/sim_data/
sim_results_MLE_spatial_20260322_151030.rds	results/sim_data/

All simulation data files are located in: `results/sim_data/`

Code modules: `code/01_spatial_setup.R` through `code/08_design_recommendations.R`

Reports: `code/00_mathematical_specification.Rmd`, `code/07_results_summary.qmd`, `code/09_MLE_design_recommendation_report.qmd`